Decentralized Indoor Direct Localization With Multiple Wi-Fi Access Points

Ziqiang Wang, *Member, IEEE*, Bo Tan, *Member, IEEE*, Mikko Valkama, *Fellow, IEEE*, Lei Xie, *Member, IEEE*, and Qun Wan, *Senior Member, IEEE*

***Abstract*—In this paper, a decentralized iterative maximum likelihood (ML) direct position determination (DIM-DPD) algorithm is proposed based on the expectation maximization (EM) concept for user equipment (UE) localization in Wi-Fi systems. By innovatively treating the non-line-of-sight (NLoS) angles-of-arrival (AoAs) and observed times-of-arrival (ToAs) as nuisance parameters in the received signal model and parameter estimation procedure, the proposed DIM-DPD demonstrates its adaptability and localization efficiency in dense multipath indoor environments. In the proposed method, the position of the UE is incorporated in the vectors denoting the location differences between the UE and the access point (APs), referred to as the UE-AP location difference vectors. The set of UE-AP location difference vectors allows constructing the related UE-AP variables, serving as the latent variables in the EM iterations. Then, by taking advantage of the alternating projection technique, the nuisance parameters and UE-AP variables in the DIM-DPD algorithm are alternatively updated on separated APs in parallel. Furthermore, instead of traditional grid search, the UE position is updated with an efficient closed-form solution by aggregating the distributed estimated low-dimensional UE-AP variables. Thus, overall, the proposed DIM-DPD approach facilitates decentralized direct localization with implementation feasible processing complexity. The provided numerical simulation results demonstrate that the proposed DIM-DPD algorithm achieves high positioning accuracy, fast convergence, and a good balance between computational complexity and performance.**

***Index Terms*—Direct position determination (DPD), expectation maximization (EM), decentralization, alternating projection, multipath, indoor localization, Wi-Fi.**

I. Introduction

WIRELESS localization and sensing will play an increasingly important role in many emerging applications, from the coordination of connected vehicles and drones in outdoor space to the navigation of personnel and robots in buildings and other indoor environments [1], [2], [3], [4]. The activation of wideband and large aperture millimeter wave (mmWave) transceivers provide large expectations on precise range and angular resolutions [5], [6]. However, factors such as high cost [7] and intermittent blockage caused by narrow beam coverage [8] may hinder the popularity of the mmWave localization systems. Therefore, the Wi-Fi-based 2.4 or 5.8 GHz localization systems remain widely used and hold also promise for further exploration [9], [10] – particularly indoors – despite the intensive studies in the past two decades [11], [12], [13]. In many existing works, the received signal strength indicator (RSSI) is a popular indicator to estimate the location of user equipment (UE) using a pathloss model [14] or utilizing fingerprint-based approaches [15]. Unfortunately, pathloss-based approaches only achieve coarse localization due to the intractable environmental factors that affect the RSSI while the fingerprint-based methods require additional server infrastructure and costly data collection campaigns. Also, both the pathloss- and fingerprint-based approaches show low adaptability to environmental changes. In addition to RSSI, the channel state information (CSI) [16], [17], [18] of the Wi-Fi devices provides another opportunity to estimate fine-grained information, for example, time-of-arrival (ToA) or time-difference-of-arrival (TDoA) for localization. However, due to the timing offsets between the digital-to-analog converter (DAC) of the access point (AP) and the analog-to-digital converter (ADC) of the UE, the direct estimation of ToA based on commodity Wi-Fi devices does not represent the signal propagation distance [19], [20]. Therefore, additional procedures, such as time-of-flight (ToF) sanitization or the Fine Timing Measurement (FTM) protocol [21], are commonly required to eliminate excessive delays and biases before using the ToA estimation for positioning.

In addition to ToA, the angle-of-arrival (AoA) is another important feature for localization due to the wide-scale adoption of multiple antennas in commodity Wi-Fi devices and the relaxed synchronization requirements between the UE and the APs. The work in [22] proved that sub-meter precision can be achieved using up to six APs that are equipped with eight-element antenna arrays in indoor multipath scenarios. In practice, it is not easy to achieve similar accuracy with commercial off-the-shelf (COTS) devices (equipped typically with three or four antenna elements), particularly in a dense indoor environment where the number of non-line-of-sight (NLoS) multipaths is difficult to resolve by a limited number of antennas. To this end, the researchers in [23], [24] and

Received 9 September 2024; revised 19 February 2025 and 16 June 2025; accepted 15 August 2025. Date of publication 29 August 2025; date of current version 22 December 2025. This work was supported in part by the Research Council of Finland (RCF) under Grant 339519, Grant 352754, Grant 357730, and Grant 359095; in part by the Business Finland under Grant 6868/31/2021; and in part by H2020 MSCA RISE under Grant 10100828. The associate editor coordinating the review of this article and approving it for publication was A. Guerra. *(Corresponding author: Bo Tan.)*

Ziqiang Wang and Mikko Valkama are with Tampere Wireless Research Center, Unit of Electrical Engineering, Tampere University, 33720 Tampere, Finland (e-mail: ziqiang.wang@tuni.fi; mikko.valkama@tuni.fi).

Bo Tan is with the Department of Electronics and Electrical Engineering, University College London, WC1E 6BT London, U.K., and also with Tampere Wireless Research Center, Unit of Electrical Engineering, Tampere University, 33720 Tampere, Finland (e-mail: bo.tan@tuni.fi; tan.bo@ucl.ac.uk).

Lei Xie is with the School of Cyber Science and Engineering, Southeast University, Nanjing 210096, China (e-mail: leixie@seu.edu.cn).

Qun Wan is with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: wanqun@uestc.edu.cn).

Digital Object Identifier 10.1109/TWC.2025.3601543

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

[25] discriminate the multipaths using joint angle and delay estimation algorithms where ToA is used as an extra dimension to separate the line-of-sight (LoS) path and the NLoS paths [19]. It should be noted that the above algorithms comprise two steps: first, to estimate intermediate parameters such as ToA and AoA on different APs; second, to aggregate intermediate parameters for position calculation. Since APs estimate intermediate parameters independently, such two-step approaches neglect the constraint that all parameters should be estimated from observations from the same signal source [26]. Extra data association will be needed to maintain performance, especially in a low signal-to-noise ratio (SNR) scenario. Therefore, the technology of *one-step direct position determination (DPD)* has drawn increasing attention recently. By modeling the source position as vector parameter in the received signal model, DPD turns the positioning into a parameter estimation problem [27].

A. State-of-the-Art in DPD Methods

DPD is a generic method for different hardware or system setups such as the number of antennas, bandwidth, and number of subcarriers. To this end, array-based DPD methods have been studied extensively [28], [29], [30]. An early study proposed a maximum likelihood DPD (ML-DPD) algorithm for a single stationary source [31], followed by a fast Fourier transform (FFT)-based implementation variant [32]. The DPD study was then extended to multiple sources in [33], which reached a balance between complexity and precision using the subspace data fusion (SDF) DPD algorithm, however, requiring prior knowledge of the number of sources. Then, a DPD algorithm derived from the distortionless minimum-variance response (MVDR) in [34] and [35] further excludes the need for the number of sources. Except for the computationally heavy and complex ML-DPD, the aforementioned methods experience a significant performance drop in the low SNR regions as a trade-off for computing efficiency. Thus, recently, importance sampling (IS) was introduced in DPD processing context in [36] and [37] to trade-off between complexity and accuracy over a wide SNR range.

Though designed for the LoS environment, DPD methods are often capable for localizing a signal source in multipath environments – however, with limited performance in the dense multipath scenario due to the limited capacity to handle the resulting energy dispersion. Thus, it becomes necessary to model the NLoS paths in the received signal for improving the DPD performance in dense multipath environments. In [38], a variant of the ML-DPD algorithm was proposed for the dense multipath scenario with the requirement of statistical multipath information. An SDF-based algorithm called multipath scenario DPD (SM-DPD) was proposed for a multipath environment in [39], with a decrease in localization accuracy as a trade-off for computational simplicity. In [40], DPD was modeled as a sparse signal recovery problem in the dense multipath environment to eliminate the need for statistical channel information in ML based DPD methods. The work was extended to the MIMO configuration in [41], however, leading to a second-order cone programming (SOCP) convex problem that results in high computational complexity. Recent work in [42] proved that grid search is an effective means of ensuring fast convergence and global optimality for sparse DPD algorithms. However, choosing the grid size is always a dilemma between complexity and accuracy. Furthermore, when summarizing DPD algorithms for dense multipath environments, one can observe that ToA (or more generally, timing information) is commonly neglected as an ingredient in the received signal models.

In addition to accuracy and complexity concerns, the possible implementation constraints form another factor that hinders the popularity of DPD. To date, most DPD methods require *centralized implementation* due to the need to aggregate radio signal samples from different APs to execute estimation and other processing algorithms. This requirement imposes additional investments in the servers and network infrastructure while also resulting into extra network traffic to convey data from the APs to the central server. Even with the advantages of high accuracy and suitability for dense multipath environments, high cost and inefficient network utilization still hinder the popularity of DPD methods [43], [44] in practice.

B. Proposed Decentralized Maximum Likelihood DPD Method

In this paper, motivated by the above theoretical and practical analysis of state-of-the-art DPD methods, we propose a *decentralized DPD algorithm with modest computational load and adaptivity in high-density multipath environments*. The considered scenario includes a single stationary UE under position estimation and multiple asynchronous stationary commercial off-the-shelf (COTS) Wi-Fi APs with the typical 4-antenna configuration (as defined in IEEE 802.11ac standard). To this end, a decentralized iterative ML DPD algorithm (DIM-DPD) is proposed based on the iterative expectation maximization (EM) technique [45]. In particular, the position of the UE is incorporated into the specific UE-AP location difference vectors based on which the UE-AP variables are constructed as latent variables and estimated using the Laplace approximation technique. Thus, a specific UE-AP variable and the corresponding nuisance parameters, i.e., AoAs of the NLoS paths and the observed ToAs of all paths, can be estimated by multiple line searches with the alternating projection technique, and further implemented in parallel in each AP. That is, the traditional multi-dimensional search involving grid points is avoided.

Furthermore, the estimation of the UE position is updated via the aggregation of low-dimensional UE-AP variables, allowing for a closed-from solution rather than grid searching method. Thus, the aggregation and processing are simple enough to be implemented in light-weight UE itself rather than at an extra central server. Finally, k-means clustering and multiple initializations are used to ensure the global optimality of final position estimation. The provided simulation results demonstrate that the proposed DIM-DPD outperforms the existing baseline solvers even in decentralized manner by comprising the observed ToA in the received signal model and by harnessing the proposed parameter estimating procedure.

In general, the proposed decentralized DIM-DPD approach differentiates from the other existing DPD methods in the following aspects: (*i*) The method is free from the requirement of any dedicated server for aggregating intermediate estimates and for final position determination; (*ii*) The method allows saving network traffic which is classically used for transmitting the raw samples between the central server and APs; (*iii*) The method avoids leaking or disclosing the position information which is usually consider private and sensitive information, by using directly the UE rather than any dedicated and external server to determine the UE position.

The contributions, novelty and advantages of the methods proposed in this paper can be summarized as follows:

1) A computationally efficient iterative DPD algorithm is proposed for UE positioning in dense multipath indoor scenarios. On the AP side, the efficiency is achieved by multiple line searching which is activated by alternating projection technique. On the UE side, a closed-form UE-AP variable aggregation and estimation method is proposed and deployed to avoid the burdensome position grid search for the final determination of the UE position.
2) Decentralized DPD implementation is manifested by two enablers. First, all searches can be implemented on separate APs in parallel by choosing UE-AP location difference vectors to construct latent variables in the expectation step of EM iteration. Second, closed-form aggregation of low-dimension UE-AP vectors (rather than complete CSI data) significantly reduces the computational load and network traffic. This fact allows avoiding centralized server processing and the related burden on the network, both being notable challenges in traditional DPD approaches, thus allowing the light-weight UE to directly execute the final position determination.
3) From a performance perspective, the proposed DIM-DPD is shown to achieve the Cramér-Rao lower bound (CRLB, derived in closed form in the Appendices) over a wide SNR range. In addition, it is shown that incorporating the observed ToAs as parameters into the received signal model and EM iteration can improve the accuracy of position estimation, despite the involved clock biases. This stems from the enhanced ability to differentiate between the involved multipath components.

The rest of this paper is organized as follows: Section II presents the signal model and problem formulation of indoor DPD with APs and dense multipath. The proposed algorithms are derived and described in Section III, where the main contributions of the paper are given. In Section IV, we analyze and discuss the properties of the proposed DPD methods. Numerical simulation results are presented and discussed in Section V. Finally, conclusions are drawn in Section VI. Selected derivations related to the involved CRLBs are provided in the Appendices.

TABLE I
MOST IMPORTANT QUANTITIES AND VARIABLES

Parameters	Meanings
$\mathbf{p}_0 \in \mathbb{R}^2$	UE position
L	The number of APs
M	The number of antennas in one AP
N	The number of subcarriers
$\mathbf{u}_l \in \mathbb{R}^2$	The position of l-th AP
Q_l	The number of NLoS paths to l-th AP
$\mathbf{a}_l \in \mathbb{C}^M$	The array steering manifold of l-th AP
$\bar{\mathbf{r}}_l \in \mathbb{C}^{MN}$	CSI vector extracted from l-th AP
$\mathbf{t}_l \in \mathbb{R}^{Q_l+1}$	observed ToAs of all paths to l-th AP
$\tilde{\boldsymbol{\theta}}_l \in \mathbb{R}^{Q_l}$	AoAs of NLoS paths to l-th AP
$\boldsymbol{\beta}_l \in \mathbb{R}^2$	l-th UE-AP location difference vector
$\boldsymbol{\mu}_l \in \mathbb{R}^2$	l-th UE-AP variable
$\tilde{\mathbf{f}}_l \in \mathbb{C}^{MN}$	The space-frequency response of one path on l-th AP
$\tilde{\mathbf{F}}_l \in \mathbb{C}^{MN\times Q_l}$	The space-frequency response contributed by NLoS paths on l-th AP

C. Definition of Notations

The Euclidean norm of any vector is denoted as $\|\cdot\|$. Moreover, $(\cdot)^T$, $(\cdot)^H$, $(\cdot)^c$, $(\cdot)^{-1}$, $\mathrm{Re}\{\cdot\}$ and $\mathrm{Im}\{\cdot\}$ denote the transpose, Hermitian transpose, conjugate, inverse and real and imaginary part operators, respectively. $|\cdot|$ stands for the absolute value, $\otimes$ denotes the Kronecker product and $\odot$ denotes the Hadamard product. $\mathbf{I}_{M_0}$ denotes the $M_0 \times M_0$ identity matrix, $\mathbf{1}_{M_0}$ is an $M_0 \times 1$ vector of ones. For a vector $\boldsymbol{\nu}$, $[\boldsymbol{\nu}]_g$ stands for g-th element in $\boldsymbol{\nu}$, $\mathrm{diag}\{\boldsymbol{\nu}\}$ is a diagonal matrix with the elements of $\boldsymbol{\nu}$ on its diagonal. For a matrix $\mathbf{X}$, $[\mathbf{X}]_{g,h}$ and $[\mathbf{X}]_g$ denote its (g,h)-th element and g-th column, respectively. The notation $\mathrm{blkdig}\{\mathbf{X}_1,\ldots,\mathbf{X}_{N_0}\}$ represents a block diagonal matrix with $\{\mathbf{X}_1,\ldots,\mathbf{X}_{N_0}\}$ on the main diagonal. $\equiv$ denotes identity and $\propto$ stands for proportionality. Finally, j is the pure complex number that verifies $j^2=-1$, and $\triangleq$ is used for definitions.

For the sake of readability and clarity, we summarize the dimensionality and brief meanings of the most important variables used in the subsequent sections in Table I. Additionally, for presentation clarity, we state that the phrasing of UE-AP refers to quantities involving both the UE and the AP(s).

II. SIGNAL MODELS AND PROBLEM FORMULATION

As shown in Fig. 1, we consider a multipath two-dimensional (2)-D) scenario with stationary UE and L static COTS APs, where the $\mathbf{p}_0 \in \mathbb{R}^2$ and $\mathbf{u}_l \in \mathbb{R}^2$ denote the position vectors of UE and l-th AP, respectively. Each AP is equipped with a uniform linear array (ULA) of M antennas with d referring to the adjacent antenna element spacing. UE is assumed to transmit the narrowband orthogonal frequency division multiplexing (OFDM) signal, for which the number of subcarriers is N, the bandwidth is B, and the carrier frequency is f_c. After deconvolution with the known pilot, the CSI vector obtained by the l-th AP is given by

$$\mathbf{r}_l(n) = b_{l0}\mathbf{a}_l(\mathbf{p}_0)e^{-j2\pi n\Delta f[t_l(\mathbf{p}_0)+\tau_{\Delta_l}]} + \sum_{q_l=1}^{Q_l} b_{lq_l}\mathbf{a}_l(\tilde{\theta}_{lq_l}) \times e^{-j2\pi n\Delta f[t_{lq_l}+\tau_{\Delta_l}]} + \mathbf{w}_l(n), \quad (1)$$

where b_{l0} and $\{b_{lq_l}\}_{q_l=1}^{Q_l}$ are complex attenuation factors. $\{\tilde{\theta}_{lq_l}\}_{q_l=1}^{Q_l}$ are the AoAs of NLoS paths; $t_l(\mathbf{p}_0)$ and t_{lq_l} denote

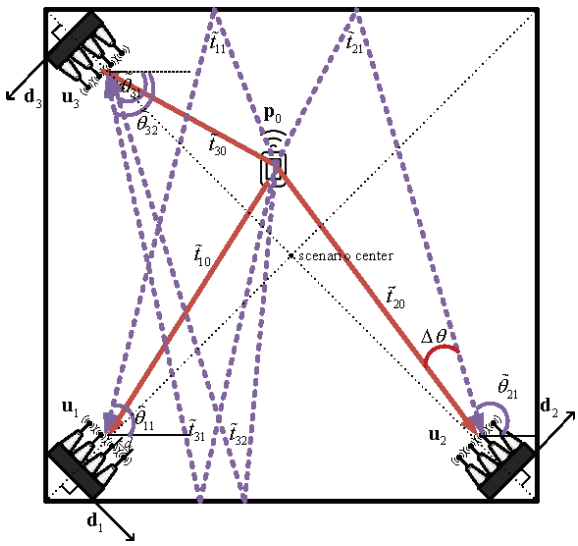


Fig. 1. Illustration of the considered multipath indoor localization scenario consisting of one UE to be located and multiple APs equipped with ULAs.

the propagation delays of LoS path and q_l-th NLoS path on l-th AP; τ_{Δ_l} reflects the random sampling offset or clock bias caused by the asynchronous nature between 802.11 APs. For notational simplicity, the time of departure (ToD) is treated as zero for all paths since there is only one UE.

$\Delta f = B/N$ is the subcarrier spacing and $n = \{0,\ldots,N-1\}$. The noise $\mathbf{w}_l(n)$ is independent of the source signal and follows the complex normal distribution $\mathcal{CN}(0,\sigma^2)$. Finally, by defining the l-th UE-AP location difference vector $\boldsymbol{\beta}_l(\mathbf{p}_0) \triangleq \mathbf{p}_0 - \mathbf{u}_l$, the time delay (TD) $t_l(\mathbf{p}_0)$ and the $(m+1)$-th element ($m = \{0,\ldots,M-1\}$) of the array manifold vector $\mathbf{a}_l$ can be given, respectively, by

$$t_l(\mathbf{p}_0) = \|\boldsymbol{\beta}_l(\mathbf{p}_0)\|/c_0,$$
$$[\mathbf{a}_l]_{m+1} = \begin{cases} e^{-j2\pi\frac{d}{\lambda}m\boldsymbol{\beta}_l^T(\mathbf{p}_0)\mathbf{d}_l/\|\boldsymbol{\beta}_l(\mathbf{p}_0)\|} & \text{operand is } \mathbf{p}_0 \\ e^{-j2\pi\frac{d}{\lambda}m[\cos(\tilde{\theta}_{lq_l}),\sin(\tilde{\theta}_{lq_l})]\mathbf{d}_l} & \text{operand is } \tilde{\theta}_{lq_l} \end{cases} \tag{2}$$

where c_0 is the signal's propagation speed, $\mathbf{d}_l$ is the known array orientation unit vector of the l-th AP and $\lambda \triangleq c_0/f_c$ is the carrier wavelength.

Note that $t_l(\mathbf{p}_0)$, t_{lq_l} and τ_{Δ_l} are unknown. Therefore, the estimation of them can be replaced by estimating sums like $\tilde{t}_{l0} \triangleq [t_l(\mathbf{p}_0) + \tau_{\Delta_l}]$ and q_l-th $\tilde{t}_{lq_l} \triangleq [t_{lq_l} + \tau_{\Delta_l}]$ that are observed ToAs of LoS and NLoS paths on the l-th AP, respectively. This fact indicates that the observed ToA here cannot be directly used for position estimation. All $\{\tilde{t}_{lq_l}\}_{q_l=1}^{Q_l}$ and $\tilde{t}_{l0}$ are considered within the Rayleigh resolution limit, which is tricky to process yet common in practical scenarios. Then, the received CSI (1) can be written in a long vector form as

$$\bar{\mathbf{r}}_l = b_{l0}\mathbf{f}_l + \tilde{\mathbf{F}}_l\tilde{\mathbf{b}}_l + \bar{\mathbf{w}}_l, \tag{3}$$

where

$$\bar{\mathbf{r}}_l \triangleq \left[\mathbf{r}_l^T(0),\ldots,\mathbf{r}_l^T(N-1)\right]^T,$$
$$\mathbf{f}_l \triangleq \mathbf{h}_l(\tilde{t}_{l0}) \otimes \mathbf{a}_l(\mathbf{p}_0),$$
$$\tilde{\mathbf{F}}_l \triangleq [\bar{\mathbf{f}}_l(\tilde{t}_{l1},\tilde{\theta}_{l1}),\ldots,\bar{\mathbf{f}}_l(\tilde{t}_{lQ_l},\tilde{\theta}_{lQ_l})],$$
$$\tilde{\mathbf{b}}_l \triangleq [b_{l1},\ldots,b_{lQ_l}]^T,$$
$$\bar{\mathbf{w}}_l \triangleq \left[\mathbf{w}_l^T(0),\ldots,\mathbf{w}_l^T(N-1)\right]^T,$$
$$\bar{\mathbf{f}}_l(\tilde{t}_{lq_l},\tilde{\theta}_{lq_l}) \triangleq \mathbf{h}_l(\tilde{t}_{lq_l}) \otimes \mathbf{a}_l(\tilde{\theta}_{lq_l}),$$
$$\mathbf{h}_l(\tilde{t}_{lq_l}) = [1, e^{-j2\pi\Delta f\tilde{t}_{lq_l}},\ldots,e^{-j2\pi(N-1)\Delta f\tilde{t}_{lq_l}}]^T. \tag{4}$$

Note that $\mathbf{h}_l(\tilde{t}_{l0})$ has the same form as $\mathbf{h}_l(\tilde{t}_{lq_l})$ based on $\tilde{t}_{l0}$. Therefore, the problem now is to estimate the location of the UE $\mathbf{p}_0$ from all CSIs $\{\bar{\mathbf{r}}_l\}_{l=1}^L$. From (3) it can be seen that except for the user location $\mathbf{p}_0$, other parameters can also be written in a vector form

$$\bar{\mathbf{t}} = [\mathbf{t}_1^T,\mathbf{t}_2^T,\ldots,\mathbf{t}_L^T]^T, \bar{\boldsymbol{\theta}} = [\tilde{\boldsymbol{\theta}}_1^T,\tilde{\boldsymbol{\theta}}_2^T,\ldots,\tilde{\boldsymbol{\theta}}_L^T]^T,$$
$$\bar{\mathbf{b}} = [\bar{\mathbf{b}}_1^T,\bar{\mathbf{b}}_2^T\ldots,\bar{\mathbf{b}}_L^T]^T = [b_{10},\tilde{\mathbf{b}}_1^T,b_{20},\tilde{\mathbf{b}}_2^T,\ldots,b_{L0},\tilde{\mathbf{b}}_L^T]^T. \tag{5}$$

where $\mathbf{t}_l = [\tilde{t}_{l0},\tilde{t}_{l1},\ldots,\tilde{t}_{lQ_l}]^T$ and $\tilde{\boldsymbol{\theta}}_l = [\tilde{\theta}_{l1},\tilde{\theta}_{l2},\ldots,\tilde{\theta}_{lQ_l}]^T$. Thus, the completely unknown parameter vector can be expressed as $\boldsymbol{\xi} \triangleq [\mathbf{p}_0^T,\bar{\mathbf{t}}^T,\bar{\boldsymbol{\theta}}^T,\bar{\mathbf{b}}^T]^T$. Then we focus on the ML estimation of $\boldsymbol{\xi}$, whose ML function is given by

$$p(\{\bar{\mathbf{r}}_l\}_{l=1}^L;\boldsymbol{\xi}) = \prod_{l=1}^L p_{\bar{\mathbf{r}}_l}(\bar{\mathbf{r}}_l;\mathbf{p}_0,\boldsymbol{\varphi}_l), \tag{6}$$

where $\boldsymbol{\varphi}_l \triangleq [\mathbf{t}_l^T,\tilde{\boldsymbol{\theta}}_l^T,b_{l0},\tilde{\mathbf{b}}_l^T]^T$ is the *nuisance parameter* vector since only UE location $\mathbf{p}_0$ is the parameter of our interest. Note that the ML function of $\boldsymbol{\xi}$ is expressed as the multiplication of multiple probability density functions (PDFs) since $\{\bar{\mathbf{r}}_l\}_{l=1}^L$ are mutually independent, and the l-th PDF is given by

$$p_{\bar{\mathbf{r}}_l}(\bar{\mathbf{r}}_l;\mathbf{p}_0,\boldsymbol{\varphi}_l) = \frac{e^{-\frac{1}{\sigma^2}\|\bar{\mathbf{r}}_l - b_{l0}\mathbf{f}_l(\mathbf{p}_0) - \tilde{\mathbf{F}}_l\tilde{\mathbf{b}}_l\|^2}}{(\pi\sigma^2)^{MN}}. \tag{7}$$

The expression in (7) indicates that in the conventional implementation for the ML estimation of $\boldsymbol{\xi}$, for each candidate position, the calculation of (6) is followed by the multi-dimensional search of $\{\boldsymbol{\varphi}_l\}_{l=1}^L$. Such approach is computationally extremely heavy to execute, even if $\bar{\mathbf{b}}$ could be replaced by closed-form expression relating $\mathbf{p}_0$ and $\boldsymbol{\varphi}_l$ (Equation (6) in [46]). To this end, we resort to the EM approach to alternatively estimate $\mathbf{p}_0$ and the nuisance parameters.

III. The Proposed Decentralized Expectation-Maximization Approach

In general, EM is a technique to iteratively estimate unknown parameters from the observation data by constructing hidden data (or latent variables). In this work and article, the observation data contains the received signal CSI at the L APs, $\mathbf{R} \triangleq \{\bar{\mathbf{r}}_l\}_{l=1}^L$; the hidden data (latent variables) are constructed using the UE-AP variables $\mathbf{U} \triangleq \{\boldsymbol{\mu}_l\}_{l=1}^L$ defined in (9); the complete unknown parameter vector under estimation is $\boldsymbol{\xi} = [\mathbf{p}_0^T,\bar{\mathbf{t}}^T,\bar{\boldsymbol{\theta}}^T,\bar{\mathbf{b}}^T]^T$ which contains the UE position $\mathbf{p}_0$ and the nuisance parameters $\boldsymbol{\varphi}_l = [\mathbf{t}_l^T,\tilde{\boldsymbol{\theta}}_l^T,b_{l0},\tilde{\mathbf{b}}_l^T]^T$. The EM iterations [47] for the estimation of all the unknown parameters in $\boldsymbol{\xi}$ can be expressed as

$$Estep\text{: Evaluate } E\left\{\log p_{\mathbf{U},\mathbf{R}}(\mathbf{U},\mathbf{R};\boldsymbol{\xi})\right\},$$
$$Mstep\text{: Solve } \boldsymbol{\xi}^{i+1} = \arg\max_{\boldsymbol{\xi}} E\left\{\log p_{\mathbf{U},\mathbf{R}}(\mathbf{U},\mathbf{R};\boldsymbol{\xi})\right\}, \tag{8}$$

where $p_{\mathbf{U},\mathbf{R}}(\cdot)$ is the PDF of joint distribution of $\mathbf{U}$ and $\mathbf{R}$, and $E \equiv E_{(\mathbf{U}|\mathbf{R};\boldsymbol{\xi}^i)}$ denotes the expectation with respect to the conditional PDF $p_{\mathbf{U}|\mathbf{R}}(\mathbf{U}|\mathbf{R};\boldsymbol{\xi}^i)$ of $\mathbf{U}$ given $\mathbf{R}$, while $\boldsymbol{\xi}^i$ is an arbitrary yet given value of the parameter vector $\boldsymbol{\xi}$ in the i-th iteration.

In this section, we propose and describe a fast and decentralized interaction method for the above EM processing referred to as the DIM-DPD approach. The general procedure of the DIM-DPD algorithm can be stated as follows:

1) We first introduce the method of separating the estimation of the UE position $\mathbf{p}_0$ and that of the other nuisance parameters in Section III-A;
2) The *Estep* and *Mstep* are then introduced and described in Sections III-B and III-C, respectively;
3) Section III-D describes the *Estep* acceleration method which is enabled by alternating projection technique [48];
4) In Section III-E, a k-means clustering based multiple initialization strategy is introduced and adopted to reduce the possibility of the proposed decentralized EM iteration being trapped in a local maxima.

A. Separation of the UE Position and Nuisance Parameters

To separate the estimation of $\mathbf{p}_0$ and nuisance parameters, we construct UE-AP variables $\{\boldsymbol{\mu}_l\}_{l=1}^L$ building on vectors $\{\boldsymbol{\beta}_l(\mathbf{p}_0)\}_{l=1}^L$ as the hidden data in our scenario, i.e.,

$$p\left(\{\boldsymbol{\mu}_l\}_{l=1}^L\right) \propto \prod_{l=1}^L e^{-\frac{1}{2}\sigma_0^{-2}\|\boldsymbol{\mu}_l-\boldsymbol{\beta}_l(\mathbf{p}_0)\|^2}, \tag{9}$$

where $\boldsymbol{\mu}_l$ is normally distributed with mean vector $\boldsymbol{\beta}_l(\mathbf{p}_0)$ and covariance matrix $\sigma_0^2\mathbf{I}_2$.

With $\{\bar{\mathbf{r}}_l\}_{l=1}^L$ and $\{\boldsymbol{\mu}_l\}_{l=1}^L$, the EM cycle for our scenario can then be expressed as

$$\begin{aligned} &Estep: \text{Evaluate} \quad \mathcal{L}\left(\boldsymbol{\xi},\boldsymbol{\xi}'\right), \\ &Mstep: \text{Solve} \quad \arg\max_{\boldsymbol{\xi}} \mathcal{L}\left(\boldsymbol{\xi},\boldsymbol{\xi}'\right), \end{aligned} \tag{10}$$

where $\boldsymbol{\xi}'$ is an value-known vector of $\boldsymbol{\xi}$ and

$$\mathcal{L}(\boldsymbol{\xi},\boldsymbol{\xi}') \overset{\Delta}{=} E_{(\{\boldsymbol{\mu}_l\}_{l=1}^L|\{\bar{\mathbf{r}}_l\}_{l=1}^L;\boldsymbol{\xi}')}\{\log p(\{\boldsymbol{\mu}_l\}_{l=1}^L,\{\bar{\mathbf{r}}_l\}_{l=1}^L;\boldsymbol{\xi})\}, \tag{11}$$

Let next $E_{\boldsymbol{\xi}'} \equiv E_{(\{\boldsymbol{\mu}_l\}_{l=1}^L|\{\bar{\mathbf{r}}_l\}_{l=1}^L;\boldsymbol{\xi}')}$ for presentation brevity. Then $\mathcal{L}(\boldsymbol{\xi},\boldsymbol{\xi}')$ in (11) can be rewritten as

$$\begin{aligned} \mathcal{L}(\boldsymbol{\xi},\boldsymbol{\xi}') &= E_{\boldsymbol{\xi}'}\{\log p(\mathbf{U})p(\mathbf{R}|\mathbf{U})\}, \\ &= E_{\boldsymbol{\xi}'}\{\log p(\{\boldsymbol{\mu}_l\}_{l=1}^L)\} + E_{\boldsymbol{\xi}'}\{\log p(\mathbf{R}|\mathbf{U})\}, \end{aligned} \tag{12}$$

where $p\left(\{\boldsymbol{\mu}_l\}_{l=1}^L\right)$ is regarded as the prior distribution, and its corresponding conditional PDF is given by

$$\begin{aligned} p(\mathbf{R}|\mathbf{U}) &= \prod_{l=1}^L p(\bar{\mathbf{r}}_l|\boldsymbol{\mu}_l;\mathbf{t}_l,\tilde{\boldsymbol{\theta}}_l,b_{l0},\tilde{\mathbf{b}}_l), \\ &\propto \prod_{l=1}^L e^{-\sigma^{-2}\|\bar{\mathbf{r}}_l-b_{l0}\mathbf{f}_l(\boldsymbol{\mu}_l)-\tilde{\mathbf{F}}_l\tilde{\mathbf{b}}_l\|^2}, \end{aligned} \tag{13}$$

where in $\mathbf{f}_l(\boldsymbol{\mu}_l)$ all $\boldsymbol{\beta}_l(\mathbf{p}_0)$ in $\mathbf{f}_l(\mathbf{p}_0)$ are replaced by $\boldsymbol{\mu}_l$. Based on (9) and (13), the EM cycle in (10) is thus equivalent to

$$\begin{aligned} &Estep: \text{Evaluate} \quad \mathcal{L}_1(\mathbf{p}_0) \text{ and } \mathcal{L}_2(\{\boldsymbol{\varphi}_l\}_{l=1}^L) \\ &Mstep: \text{Solve } \arg\max_{\mathbf{p}_0} \mathcal{L}_1(\mathbf{p}_0) \text{ and } \arg\max_{\{\boldsymbol{\varphi}_l\}_{l=1}^L} \mathcal{L}_2(\{\boldsymbol{\varphi}_l\}_{l=1}^L), \end{aligned} \tag{14}$$

where

$$\mathcal{L}_1(\mathbf{p}_0) = -\frac{1}{2}\sigma_0^{-2}\sum_{l=1}^L E_{\boldsymbol{\xi}'}\|\boldsymbol{\mu}_l-\boldsymbol{\beta}_l(\mathbf{p}_0)\|^2, \tag{15a}$$

$$\mathcal{L}_2(\{\boldsymbol{\varphi}_l\}_{l=1}^L) = -\sigma^{-2}\sum_{l=1}^L E_{\boldsymbol{\xi}'}\|\bar{\mathbf{r}}_l-b_{l0}\mathbf{f}_l(\boldsymbol{\mu}_l)-\tilde{\mathbf{F}}_l\tilde{\mathbf{b}}_l\|^2. \tag{15b}$$

It can be observed from (14) and (15) that the estimation of $\mathbf{p}_0$ and that of the nuisance parameters are separated. Next, we derive the solutions for the *EStep* and *MStep* in (14), respectively.

B. Estep

Estep in (14) is equivalent to estimating the conditional expectation $E_{\boldsymbol{\xi}'}\{\boldsymbol{\mu}_l\}$ and $E_{\boldsymbol{\xi}'}\{\mathbf{f}_l(\boldsymbol{\mu}_l)\}$ because random variables in (15a) and (15b) are, respectively, $\boldsymbol{\mu}_l$ and $\mathbf{f}_l(\boldsymbol{\mu}_l)$.

For the estimation of both, we recall the following lemma [49]:

Lemma 1: Consider the multivariate function $\mathbf{g}(\boldsymbol{\upsilon}) : \mathbb{R}^{N_0} \to \mathbb{R}^{M_0}$, $f(\boldsymbol{\upsilon}) : \mathbb{R}^{N_0} \to \mathbb{R}$ and the integral $I_f = \int_{\boldsymbol{\upsilon}\in\mathbf{V}} \mathbf{g}(\boldsymbol{\upsilon})\, e^{f(\boldsymbol{\upsilon})} d\boldsymbol{\upsilon}$. Supposing that $f(\boldsymbol{\upsilon})$ has a global maximum strictly within the interval $\mathbf{V}$, the Laplace approximation for I_f is given by

$$\begin{aligned} I_f &\approx \mathbf{g}(\boldsymbol{\upsilon}^*)\, e^{f(\boldsymbol{\upsilon}^*)} \int\cdots\int e^{\frac{1}{2}(\boldsymbol{\upsilon}-\boldsymbol{\upsilon}^*)^T\mathbf{F}(\boldsymbol{\upsilon}^*)(\boldsymbol{\upsilon}-\boldsymbol{\upsilon}^*)} d\boldsymbol{\upsilon}, \\ &= (2\pi)^{N_0/2}(\det[-\mathbf{H}(\boldsymbol{\upsilon}^*)])^{-1/2}\mathbf{g}(\boldsymbol{\upsilon}^*)\, e^{f(\boldsymbol{\upsilon}^*)}. \end{aligned} \tag{16}$$

where $\mathbf{H}(\boldsymbol{\upsilon}^*)$ is the Hessian matrix of $f(\boldsymbol{\upsilon})$ evaluated at the global maximum point $\boldsymbol{\upsilon}^*$ while $\det[\cdot]$ stands for the determinant operator.

Due to the fact that both $\mathbf{f}_l(\boldsymbol{\mu}_l)$ and $\boldsymbol{\mu}_l$ are only related to the l-th AP, we define $\check{\mathbf{g}}(\boldsymbol{\mu}_l) \overset{\Delta}{=} E_{(\boldsymbol{\mu}_l|\bar{\mathbf{r}}_l;\boldsymbol{\xi}')}\{\mathbf{g}(\boldsymbol{\mu}_l)\}$. For the sake of presentation brevity, the subscript l is omitted in the below derivations (17) to (20). To this end, we can now write

$$\begin{aligned} \check{\mathbf{g}}(\boldsymbol{\mu}) &= \int \mathbf{g}(\boldsymbol{\mu})\, p_{\boldsymbol{\mu}|\bar{\mathbf{r}}}\, d\boldsymbol{\mu} = \int \mathbf{g}(\boldsymbol{\mu})\frac{p_{\bar{\mathbf{r}}|\boldsymbol{\mu}}\, p_{\boldsymbol{\mu}}}{p_{\bar{\mathbf{r}}}} d\boldsymbol{\mu}, \\ &\propto \frac{1}{p_{\bar{\mathbf{r}}}}\int \mathbf{g}(\boldsymbol{\mu}) e^{-\frac{\|\bar{\mathbf{r}}-b_0\mathbf{f}-\tilde{\mathbf{F}}\tilde{\mathbf{b}}\|^2}{\sigma^2}-\frac{\|\boldsymbol{\mu}-\boldsymbol{\beta}\|^2}{2\sigma_0^2}} d\boldsymbol{\mu}, \end{aligned} \tag{17}$$

Turning next to the polar coordinate system and redefining

$$\boldsymbol{\mu} = \rho_\mu[\cos\psi_\mu, \sin\psi_\mu]^T \text{and} \boldsymbol{\beta} = \rho_\beta[\cos\psi_\beta, \sin\psi_\beta]^T, \tag{18}$$

then (17) can be rewritten as

$$\begin{aligned} \check{\mathbf{g}}(\boldsymbol{\mu}) \propto \frac{1}{p_{\bar{\mathbf{r}}}}\int_0^\infty\int_{-\pi}^{\pi} \rho_\mu e^{-\frac{\rho_\mu^2}{2\sigma_0^2}} \times \\ \mathbf{g}(\rho_\mu,\psi_\mu) e^{\frac{\rho_\beta}{\sigma_0^2}\rho_\mu\cos(\psi_\mu-\psi_\beta)-\frac{\|\bar{\mathbf{r}}-b_0\bar{\mathbf{f}}(\psi_\mu)-\tilde{\mathbf{F}}\tilde{\mathbf{b}}\|^2}{\sigma^2}} d\rho_\mu d\psi_\mu, \end{aligned} \tag{19}$$

while similarly $p_{\bar{\mathbf{r}}}$ can be given as

$$\begin{aligned} p_{\bar{\mathbf{r}}} \propto \int_0^\infty\int_{-\pi}^{\pi} \rho_\mu e^{-\frac{\rho_\mu^2}{2\sigma_0^2}} \times \\ e^{\frac{\rho_\beta}{\sigma_0^2}\rho_\mu\cos(\psi_\mu-\psi_\beta)-\frac{\|\bar{\mathbf{r}}-b_0\bar{\mathbf{f}}(\psi_\mu)-\tilde{\mathbf{F}}\tilde{\mathbf{b}}\|^2}{\sigma^2}} d\rho_\mu d\psi_\mu. \end{aligned} \tag{20}$$

Furthermore, let $\mathbf{g}(\boldsymbol{\mu}_l) = f_l(\boldsymbol{\mu}_l)$, then by applying the result of *Lemma 1*, $E_{\boldsymbol{\xi}'}\{\mathbf{f}_l(\boldsymbol{\mu}_l)\}$ can be obtained by taking the ratio

of integral terms' Laplace approximations in (19) and (20). Similarly $E_{\boldsymbol{\xi}'}\{\boldsymbol{\mu}_l\}$ can be obtained with $\mathbf{g}(\boldsymbol{\mu}_l)=\boldsymbol{\mu}_l$, i.e., let $\check{\mathbf{f}}_l \equiv E_{\boldsymbol{\xi}'}\{\mathbf{f}_l(\boldsymbol{\mu}_l)\}$ and $\check{\boldsymbol{\mu}}_l \equiv E_{\boldsymbol{\xi}'}\{\boldsymbol{\mu}_l\}$ then we have

$$\check{\mathbf{f}}_l \approx \mathbf{f}_l(\boldsymbol{\mu}_l^*) \quad \text{and} \quad \check{\boldsymbol{\mu}}_l \approx \boldsymbol{\mu}_l^*, \tag{21}$$

where $\boldsymbol{\mu}_l^* = \rho_{\mu_l}^*[\cos\psi_{\mu_l}^*, \sin\psi_{\mu_l}^*]^T$. Thus, the solution for *Estep*, can be obtained from the following maximization problem

$$\begin{aligned}[\rho_{\mu_l}^*,\psi_{\mu_l}^*]^T = &\arg\max_{[\rho_{\mu_l},\psi_{\mu_l}]^T} \eta\left[-\frac{\rho_{\mu_l}^2}{2}+\rho_{\mu_l}\rho_{\beta_l}\cos(\psi_{\mu_l}-\psi_{\beta_l})\right]\\ &-\bar{\mathcal{L}}_2^l(\psi_{\mu_l},\boldsymbol{\varphi}_l),\\ s.t. \quad &\rho_{\mu_l}>0, -\pi\le\psi_{\mu_l}<\pi.\end{aligned} \tag{22}$$

where $\eta=\sigma^2/\sigma_0^2$ can be used to control the convergence speed [47] and

$$\bar{\mathcal{L}}_2^l(\psi_{\mu_l},\boldsymbol{\varphi}_l) = \|\bar{\mathbf{r}}_l - b_{l0}\bar{\mathbf{f}}_l(\psi_{\mu_l},t_{l0}) - \tilde{\mathbf{F}}_l\tilde{\mathbf{b}}_l\|^2. \tag{23}$$

It can be seen that once ψ_{μ_l} is selected, (22) becomes a convex quadratic function of ρ_{μ_l}. Thus, the maximization of (22) can be transformed to a line search only over ψ_{μ_l}, i.e., with an arbitrary selected ψ'_{μ_l}, the corresponding ρ'_{μ_l} is given by

$$\rho'_{\mu_l} = \rho_{\beta_l}\cos\left(\psi'_{\mu_l}-\psi_{\beta_l}\right). \tag{24}$$

Then the optimization problem in (22) can be further expressed as

$$\psi_{\mu_l}^* = \arg\max_{\psi_{\mu_l}} \eta\mathcal{F}(\psi_{\mu_l}) - \bar{\mathcal{L}}_2^l(\psi_{\mu_l},\boldsymbol{\varphi}_l), \tag{25a}$$

$$\rho_{\mu_l}^* = \rho_{\beta_l}\cos(\psi_{\mu_l}^*-\psi_{\beta_l}), s.t. \rho_{\mu_l}>0, -\pi\le\psi_{\mu_l}<\pi, \tag{25b}$$

where

$$\mathcal{F}(\psi_{\mu_l}) = [\rho_{\beta_l}\cos(\psi_{\mu_l}-\psi_{\beta_l})]^2/2. \tag{26}$$

It can now be seen from (25a) that η constrains the length of the iteration step to guarantee the fast enough convergence. The larger η improves the effect of $\mathcal{F}(\psi_{\mu_l})$, resulting in a smaller Euclidean distance between updated $\hat{\mathbf{p}}_0$ and last $\hat{\mathbf{p}}_0$ in terms of the polar angle. Actually, even if there is no $\mathcal{F}(\psi_{\mu_l})$, (25b) can still provide a constraint on the iteration step in terms of radial distance ρ_{μ_l}. Thus, giving up $\mathcal{F}_l(\psi_{\mu_l})$ to relax the constraint in *Estep*. Then η can be set as small as possible, that is, $\eta\to 0$. Thus, the parameter design process is eventually essentially avoided, which contributes to the universality and deployment simplicity of the proposed algorithm.

C. Mstep

With $\{\boldsymbol{\mu}_l^*\}_{l=1}^L$, the maximization of (15a) can be rewritten as

$$\arg\max_{\mathbf{p}_0} -\sum_{l=1}^{L}\|\boldsymbol{\mu}_l^* - \mathbf{p}_0 + \mathbf{u}_l\|^2, \tag{27}$$

which easily yields

$$\hat{\mathbf{p}}_0 = \frac{1}{L}\sum_{l=1}^{L}(\mathbf{u}_l+\boldsymbol{\mu}_l^*). \tag{28}$$

Similarly, we can transform the maximization of (15b) as

$$\{\hat{\boldsymbol{\varphi}}_l\}_{l=1}^L = \arg\max_{\{\boldsymbol{\varphi}_l\}_{l=1}^L}\sum_{l=1}^{L} -\bar{\mathcal{L}}_2^l(\psi_{\mu_l}^*,\boldsymbol{\varphi}_l). \tag{29}$$

Equation (29) indicates that all $\{\boldsymbol{\varphi}_l\}_{l=1}^L$ can be estimated on their corresponding APs separately. For the sake of brevity, we only focus on estimating the l-th one of $\{\boldsymbol{\varphi}_l\}_{l=1}^L$, which can be easily extended to other ones. Thus, the following problem is formulated

$$[\hat{\mathbf{t}}_l, \hat{\tilde{\boldsymbol{\theta}}}_l] = \arg\max_{[\mathbf{t}_l,\tilde{\boldsymbol{\theta}}_l]} \bar{\mathcal{L}}_c^l(\psi_{\mu_l}^*,\mathbf{t}_l,\tilde{\boldsymbol{\theta}}_l), \tag{30}$$

where $\bar{\mathcal{L}}_c^l(\psi_{\mu_l}^*,\mathbf{t}_l,\tilde{\boldsymbol{\theta}}_l)$ is a concise version of $\bar{\mathcal{L}}_2^l(\psi_{\mu_l}^*,\boldsymbol{\varphi}_l)$ by eliminating b_{l0} and $\tilde{\mathbf{b}}_l$ [46], which is given by

$$\bar{\mathcal{L}}_c^l(\psi_{\mu_l}^*,\mathbf{t}_l,\tilde{\boldsymbol{\theta}}_l) = \bar{\mathbf{r}}_l^H\bar{\mathbf{F}}_l\bar{\mathbf{D}}_l^{-1}\bar{\mathbf{F}}_l^H\bar{\mathbf{r}}_l, \tag{31}$$

where $\bar{\mathbf{D}}_l = \bar{\mathbf{F}}_l^H\bar{\mathbf{F}}_l$ and

$$\bar{\mathbf{F}}_l \equiv \bar{\mathbf{K}}([\mathbf{t}_l,[\psi_{\mu_l}^*,\tilde{\boldsymbol{\theta}}_l^T]^T]) \triangleq [\bar{\mathbf{f}}(\psi_{\mu_l}^*,t_{l0}),\tilde{\mathbf{F}}_l]. \tag{32}$$

As $\mathbf{e}_l \triangleq [\mathbf{t}_l^T,\tilde{\boldsymbol{\theta}}_l^T]^T \in \mathbb{R}^{2Q_l+1}$ is a multi-dimensional vector, estimating it by exhaustive search is computationally heavy and impractical. To achieve a tradeoff between computational load and estimation accuracy, we consider the implementation that alternatively estimates each element in $\mathbf{e}_l$ when fixing the values of others. That is

$$\widehat{[\mathbf{e}_l]}_{id} = \arg\max_{[\mathbf{e}_l]_{id}} \bar{\mathcal{L}}_c^l([\mathbf{e}_l]_{id},[\psi_{\mu_l}^*,(\mathbf{e}'_l)_{id}^T]^T), \tag{33}$$

where $(\mathbf{e}'_l)_{id}$ is the vector consisting of elements in value-known $\mathbf{e}'_l$ excluding $[\mathbf{e}'_l]_{id}$, while $\mathbf{e}'_l$ is updated once $\widehat{[\mathbf{e}_l]}_{id}$ is obtained. Obviously, and importantly, only the line search is required for (33), no matter how many NLoS paths exist.

The derivations of the *Estep* and the *Mstep* are now completed. Next, we focus on the efficient implementation of the EM iterations.

D. Acceleration of EM Iteration Cycle

Firstly, as estimating $\{b_{l0}\}_{l=1}^L$ and $\{\tilde{\mathbf{b}}_l\}_{l=1}^L$ is not directly necessary for optimizing the original ML function (6) [46], $\bar{\mathcal{L}}_c^l(\cdot)$ in (31) can be used to replace $\bar{\mathcal{L}}_2^l(\cdot)$ in (22). Thus, the $(i+1)$-th iteration cycle of the proposed DPD algorithm can further be re-expressed as

$$\begin{aligned}&Estep:\\ &\psi_{\mu_l}^* = \arg\max_{\psi_{\mu_l}} \bar{\mathcal{L}}_c^l(\psi_{\mu_l},\mathbf{e}_l^i), &\text{(34a)}\\ &\rho_{\mu_l}^* = \rho_{\beta_l}\cos(\psi_{\mu_l}^*-\psi_{\beta_l}), &\text{(34b)}\\ &s.t.\ \rho_{\mu_l}>0, -\pi\le\psi_{\mu_l}<\pi,\\ &Mstep:\\ &\widehat{[\mathbf{e}_l^{i+1}]}_{id} = \arg\max_{[\mathbf{e}_l]_{id}} \bar{\mathcal{L}}_c^l([\mathbf{e}_l]_{id},[\psi_\mu^*,(\mathbf{e}_l^i)_{id}^T]^T), &\text{(34c)}\\ &id=1,\ldots,2Q_l+1,\\ &\hat{\mathbf{p}}_0^{i+1} = \frac{1}{L}\sum_{l=1}^{L}\left[\mathbf{u}_l+\rho_{\mu_l}^*\begin{bmatrix}\cos\psi_{\mu_l}^*\\ \sin\psi_{\mu_l}^*\end{bmatrix}\right], &\text{(34d)}\end{aligned}$$

where $\boldsymbol{\beta}_l^i = \rho_{\beta_l^i}[\cos\psi_{\beta_l^i},\sin\psi_{\beta_l^i}]^T = \hat{\mathbf{p}}_0^i - \mathbf{u}_l$.

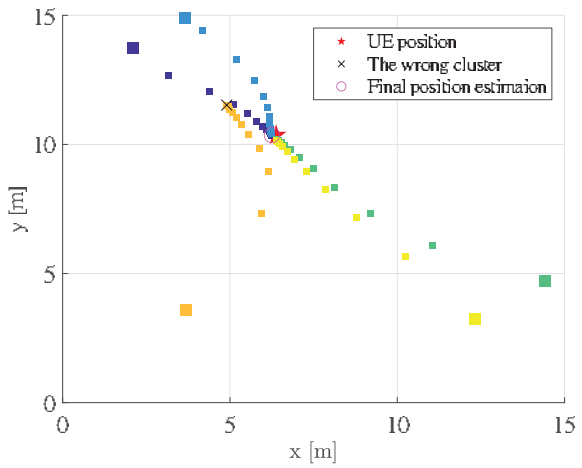


Fig. 2. Illustration of position update and final position determination in DIM-DPD.

Note that the cost function $\bar{\mathcal{L}}_c^l(\cdot)$ exists in both *Estep* (34a) and *Mstep* (34c). It can be seen that, the matrix multiplication and matrix inversion for $\bar{\mathbf{D}}_l$ is repeatedly implemented for each candidate ψ_{μ_l}. However, except elements related to ψ_{μ_l}, all other Q_l^2 ones in $\bar{\mathbf{D}}_l$ are known and unchanged in this search process. That is, there are many redundant calculations in current *Estep*. To this end, the *alternating projection* technique [48] is used to simplify $\bar{\mathcal{L}}_c^l(\psi_{\mu_l}, \mathbf{e}_l^i)$ as a function $\check{\mathcal{L}}_c^l([\mathbf{E}_l]_{1,2}, (\mathbf{E}_l^i)_{1,2})$, i.e.,

$$\check{\mathcal{L}}_c^l([\mathbf{E}_l]_{1,2},(\mathbf{E}_l^i)_{1,2})=\frac{\bar{\mathbf{f}}_l^H(\psi_{\mu_l},t_{l0}^i)\mathbf{P}^{\perp}_{(\bar{\mathbf{F}}_l^i)_1}\bar{\mathbf{r}}_l\bar{\mathbf{r}}_l^H\mathbf{P}^{\perp}_{(\bar{\mathbf{F}}_l^i)_1}\bar{\mathbf{f}}_l(\psi_{\mu_l},t_{l0}^i)}{\bar{\mathbf{f}}_l^H(\psi_{\mu_l},t_{l0}^i)\mathbf{P}^{\perp}_{(\bar{\mathbf{F}}_l^i)_1}\bar{\mathbf{f}}_l(\psi_{\mu_l},t_{l0}^i)}, \tag{35}$$

where $\mathbf{E}_l \triangleq [\mathbf{t}_l, [\psi_{\mu_l}, \tilde{\boldsymbol{\theta}}_l^T]^T] \in \mathbb{R}^{(Q_l+1)\times 2}$, $(\mathbf{E}_l^i)_{g,h}$ consists of elements in value-known $\mathbf{E}_l^i$ excluding $[\mathbf{E}_l^i]_{g,h}$, while $\mathbf{E}_l^i$ is updated once $\widehat{[\mathbf{E}_l]}_{g,h}$ is obtained by maximizing $\check{\mathcal{L}}_c^l([\mathbf{E}_l]_{g,h}, (\mathbf{E}_l^i)_{g,h})$. Let $\bar{\mathbf{F}}_l^i \equiv \bar{\mathbf{K}}(\mathbf{E}_l^i)$, then we have

$$\mathbf{P}^{\perp}_{(\bar{\mathbf{F}}_l^i)_g} = \mathbf{I}_{MN} - (\bar{\mathbf{F}}_l^i)_g[(\bar{\mathbf{F}}_l^i)_g^H(\bar{\mathbf{F}}_l^i)_g]^{-1}(\bar{\mathbf{F}}_l^i)_g^H. \tag{36}$$

where $(\bar{\mathbf{F}}_l^i)_g$ denotes $\bar{\mathbf{F}}_l^i$ excluding the g-th column. Note that the aforementioned transformation from $\bar{\mathcal{L}}_c^l(\psi_{\mu_l}, \mathbf{e}_l^i)$ to $\check{\mathcal{L}}_c^l([\mathbf{E}_l]_{1,2}, (\mathbf{E}_l^i)_{1,2})$ can be transferred to transform $\bar{\mathcal{L}}_c^l([\mathbf{e}_l]_{id}, [\psi_{\mu_l}^*, (\mathbf{e}_l^i)_{id}^T]^T)$ in (34c).

E. Multiple-Point Initialization Strategy

The proposed algorithm is iterative in nature. To avoid being trapped by local maxima as far as possible, S different initial points and *LS* initial NLoS AoA vectors are used to generate S candidate position outputs $\{\hat{\mathbf{p}}_f^s\}_{s=1}^S$, while being then further clustered using the k-means approach. The final position estimation is discriminated by: 1) belonging to the cluster with largest number of elements; and 2) being closest to the cluster centroid $\mathbf{p}_c$. That is,

$$\hat{\mathbf{p}}_0 = \arg\min_{\hat{\mathbf{p}}_f^s} \|\hat{\mathbf{p}}_f^s - \mathbf{p}_c\|, \ s = 1, 2, \ldots, S. \tag{37}$$

Fig. 2 illustrates the typical position update and the final position determination progress. As for the initialization of all ToAs, recalling that they are all within the Rayleigh resolution

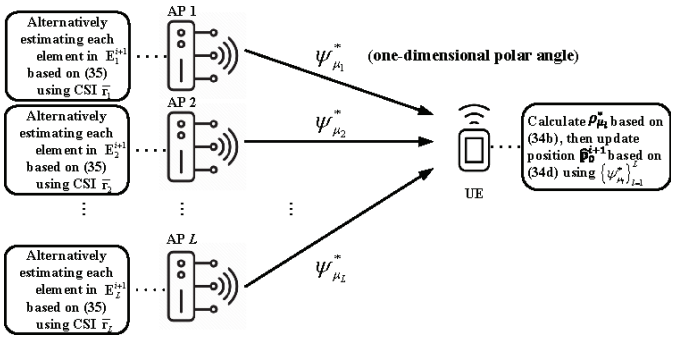


Fig. 3. Schematic diagram representing the decentralized nature of the DIM-DPD in one iteration: 1) Estimating the polar angles of UE-AP variables at distributed APs; 2) Position estimation update at UE.

TABLE II
COMMUNICATION OVERHEAD COMPARISON, ASSUMING $N = 64$, $M = 4$, $S = 5$, $N_{it} = 10$

Algorithm	Communication Overhead	Data type	NCL
DIM-DPD	SN_{it}	Real	1
CDPD	MN	Complex	5.12

limit, the solution of following single path ToA estimation problem can be used [46], expressed as

$$\hat{\tilde{t}}_l = \arg\max_{\tilde{t}_l} \sum_{m=1}^{M} |[\mathbf{R}_l]_m^H \mathbf{h}(\tilde{t}_l)|^2, \tag{38}$$

where

$$\mathbf{R}_l \triangleq [\bar{\mathbf{r}}_l(0), \ldots, \bar{\mathbf{r}}_l(N-1)]^T. \tag{39}$$

Compared with (3), (38) is derived by modeling the ToA of only one path for $\bar{\mathbf{r}}_l$, while the other paths or components are assumed to be additive noise from the initialization perspective.

For the sake of readability and presentation clarity, the overall workflow of the proposed decentralized iterative ML DPD algorithm (DIM-DPD) is summarized in Algorithm 1.

IV. PROPERTIES OF THE PROPOSED DPD ALGORITHM

A. Decentralization

As can be observed from Algorithm 1, within each iteration cycle, only the step 18 requires aggregating data from all APs for position update. Moreover, instead of collecting the complete CSI or other raw data from all APs, as in traditional DPD methods, the data that is shared in our proposed approach contains only $\{\psi_{\mu_l}^*\}_{l=1}^L$. To this end, Fig. 3 illustrates the calculation and aggregation process for the proposed DIM-DPD. Therefore, the heavy network traffic stemming from sharing and transmitting the raw data in traditional DPD algorithms is here avoided. The proposed DIM-DPD approach is thus clearly more favorable for the decentralized implementation.

To address this more explicitly, let the number of iterations be N_{it}. The corresponding communication overheads of the proposed DIM-DPD method and the centralized DPD (CDPD) are summarized in Table II. If we use the DIM-DPD communications load as the reference, the normalized communications

Algorithm 1 Summary of the Proposed Decentralized Iterative DIM-DPD Method

Input: CSIs $\{\bar{\mathbf{r}}_l\}_{l=1}^L$, the stopping threshold ϵ.

1 **for** $l \leftarrow 1$ **to** L **do**
2 $\quad \hat{\tilde{t}}_l = \arg\max_{\tilde{t}_l} \sum_{m=1}^M |[\mathbf{R}_l]_m^H \mathbf{h}(\tilde{t}_l)|^2$ (38);
3 **end**
4 **for** $s \leftarrow 1$ **to** S **do**
5 $\quad$ Choosing the initial position $\hat{\mathbf{p}}_0^0$ randomly in the area that UE is located in, let $i = 0$;
6 $\quad$ **for** $l \leftarrow 1$ **to** L **do**
7 $\quad\quad$ calculating $\psi_{\beta_l^0}$ based on $\hat{\mathbf{p}}_0^0$ and randomly choosing initial NLoS AoA vector $\tilde{\boldsymbol{\theta}}_l^0 \in \mathbb{R}^{Q_l}$;
8 $\quad\quad$ Constructing $\mathbf{E}_l^0 = [\hat{\tilde{t}}_l \cdot \mathbf{1}_{Q_l+1}, [\psi_{\beta_l^0}, (\tilde{\boldsymbol{\theta}}_l^0)^T]]$;
9 $\quad$ **end**
10 $\quad$ **repeat**
11 $\quad\quad$ **for** $l \leftarrow 1$ **to** L **do**
12 $\quad\quad\quad$ $\rho_{\beta_l^i}[\cos\psi_{\beta_l^i}, \sin\psi_{\beta_l^i}]^T = \hat{\mathbf{p}}_0^i - \mathbf{u}_l$;
13 $\quad\quad\quad$ **for** $\bar{q}_l \leftarrow 1$ **to** $Q_l + 1$ **do**
14 $\quad\quad\quad\quad$ $\widehat{[\mathbf{E}_l^{i+1}]}_{\bar{q}_l,1} = \arg\max_{[\mathbf{E}_l]_{\bar{q}_l,1}} \check{\mathcal{L}}_c^l([\mathbf{E}_l]_{\bar{q}_l,1}, (\mathbf{E}_l^i)_{\bar{q}_l,1})$ (34c);
15 $\quad\quad\quad\quad$ $\widehat{[\mathbf{E}_l^{i+1}]}_{\bar{q}_l,2} = \arg\max_{[\mathbf{E}_l]_{\bar{q}_l,2}} \check{\mathcal{L}}_c^l([\mathbf{E}_l]_{\bar{q}_l,2}, (\mathbf{E}_l^i)_{\bar{q}_l,2})$ (34c);
16 $\quad\quad\quad$ **end**
17 $\quad\quad\quad$ Rearranging $\mathbf{E}_l^{i+1}$ by rows in ascending order according to $[\mathbf{E}_l^{i+1}]_1$;
18 $\quad\quad\quad$ Aggregating $\psi_{\mu_l}^* = [\mathbf{E}_l^{i+1}]_{1,2}$ to the UE;
19 $\quad\quad\quad$ $\rho_{\mu_l}^* = \rho_{\beta_l^i} \cos\left(\psi_{\mu_l}^* - \psi_{\beta_l^i}\right)$ (34b);
20 $\quad\quad$ **end**
21 $\quad\quad$ $\hat{\mathbf{p}}_0^{i+1} = \frac{1}{L}\sum_{l=1}^L \left[\mathbf{u}_l + \rho_{\mu_l}^* \begin{bmatrix}\cos\psi_{\mu_l}^* \\ \sin\psi_{\mu_l}^*\end{bmatrix}\right]$ (34d);
22 $\quad\quad$ $i \leftarrow i + 1$;
23 $\quad$ **until** $|\hat{\mathbf{p}}_0^i - \hat{\mathbf{p}}_0^{i-1}| < \epsilon$;
24 $\quad$ $\hat{\mathbf{p}}_f^s = \hat{\mathbf{p}}_0^i$;
25 **end**
26 Clustering $\{\hat{\mathbf{p}}_f^s\}_{s=1}^S$ by k-means approach;
27 $\hat{\mathbf{p}}_0 = \arg\min_{\hat{\mathbf{p}}_f^s} \|\hat{\mathbf{p}}_f^s - \mathbf{p}_c\|$, $s = 1, 2, \ldots, S$ (37).

load (NCL) of CDPD is 5.12 times higher. As the CSI dimension *MN* increases, the computational load of CDPD becomes gradually heavier and heavier. In addition, both steps 19 and 21 involve only simple operations of $\{\psi_{\mu_l}^*\}_{l=1}^L$, thus even a light-weight UE (e.g. a mobile phone or other similar device) itself may complete the final steps of the DIM-DPD for obtaining the estimated device position. The additional central processing unit (server) is thus unnecessary.

B. Computational Complexity

Next, we assess the computation complexity of the proposed DPD algorithms and their state-of-the-art counterparts. Complexity is counted or measured via the order of the number of complex-valued multiplications. For all algorithms compared, the numbers of grid points, candidate angles and ToAs are N_g, N_d and N_t, respectively. Considering that there is no grid search in the proposed DIM-DPD, most of the computational resource is consumed by maximizing (35) in the main iteration cycle of Algorithm 1 (steps 14-15). Further recalling that (35) involves matrix inversion, the total computational complexity of the proposed algorithm is $O\left(SN_{it}(N_d+N_t)\left(\sum_{l=1}^L Q_l^3 + Q_l^2 MN\right)\right)$. Despite not using ToA estimation, another iterative DPD algorithm ADMM-DPD [44] searches N_g points to estimate position in each iteration and the corresponding computational complexity is $O(NN_{it}((LN_g)^2 + (LN_d)^2)LM)$. Regarding the subspace data fusion based SM-DPD algorithm [39] in the multipath scenario, its computational complexity is of the form $O(L(M^3 + M^2N + N_gM))$ since it requires eigenvalue decomposition of the CSI covariance matrix. The exhaustive ML grid search algorithm (EX-DPD), in turn, requires a multi-dimensional exhaustive search, resulting into impractical computational complexity of $O\left(N_g N \sum_{l=1}^L (N_d^{Q_l}(Q_l^2 M + Q_l^3))\right)$. It can be seen that when $N_g \gg N_t$, the proposed DIM-DPD method exhibits remarkable computational savings compared to the advanced ADMM-DPD. Furthermore, the condition that N_g is much larger than N_t is common in indoor scenarios, especially when the localization area is large.

TABLE III
COMPLEXITY ASSESSMENT OF THE CONSIDERED DPD ALGORITHMS, FOR $S = 5$, $N = 64$, $M = 4$, $L = 3$, $\{Q_l\}_{l=1}^L = 3$, $N_{it} = 10$, $N_g = 10^3$, $N_t = 50$, $N_d = 100$

Algorithm	Complexity	NCC
DIM-DPD	$O(SN_{it}(N_d+N_t)(\sum_{l=1}^L Q_l^3 + Q_l^2 MN))$	1
ADMM-DPD	$O(NN_{it}((LN_g)^2 + (LN_d)^2)LM)$	1331.06
SM-DPD	$O(L(M^3 + M^2N + N_gM))$	0.0002
EX-DPD	$O(N_gN\sum_{l=1}^L (N_d^{Q_l}(Q_l^2M + Q_l^3)))$	23063.06

The above complexity observations are summarized in Table III, while noting also the corresponding normalized computational complexities (NCCs) where the DIM-DPD serves as the reference. It can be clearly observed that the proposed method allows for favorable processing complexity compared to many reference works. Furthermore, and importantly, only the proposed DIM-DPD algorithm allows for decentralized processing, i.e., (35) in DIM-DPD can be implemented on each AP separately in parallel.

C. Position Estimation With Blocked APs

In practice, it is very likely that the LoS path between the UE and (some of) the APs may be completely blocked due to the existence of obstructions in the indoor environments. In this condition, using CSIs from all APs – including the blocked ones – to implement the DPD will inevitably result in severe position estimation bias. Thus, it is beneficial to detect CSIs from APs that contain only NLoS paths, and remove these from the algorithm, to ensure the high accuracy of the estimation. To this end, [44] performs detection based on the attenuation of the path. However, attenuation is not a stable

feature to discriminate the LoS path from the NLoS paths. Especially when the LoS path is not completely blocked, its attenuation can be at the same level or even less compared to the NLoS path. Instead of using path attenuation, the proposed algorithm focuses on the angular information of all the first paths. That is, inspired by the principle of DPD that all LoS paths are from the same target, among the final $\{\psi_{\mu_l^*}\}_{l=1}^{L}$ belonging to $\hat{\mathbf{p}}_0$, only the ones $\{\psi_{\mu_l^*}\}_{l'=1}^{L'}(L' \leq L)$ that can converge into one concentrated region are judged to belong to the LoS paths. Then, by implementing steps 12, 19 and 21 in Algorithm 1, $\{\psi_{\mu_l^*}\}_{l'=1}^{L'}$ and $\hat{\mathbf{p}}_0$ are used to obtain a refined position estimate $\hat{\mathbf{p}}_r$.

D. Accuracy Enhancement Through Observed ToAs

Due to the time offsets and bias between the UE and APs in 802.11 systems, observed ToA of the LoS path cannot be directly used for the positioning purpose. However, the relative relations between the observed ToAs of all multipaths remain a useful indicator to discriminate the LoS path. Hence, comprising observed ToA in the received signal model and the iterative estimation procedure will benefit the position estimation accuracy. This can be theoretically verified by the following preposition.

Proposition 1: Defining the CRLB of position block in two different cases: 1) $\mathbf{CRB}^{\mathrm{O}}_{\mathbf{p}_0\mathbf{p}_0}$ for joint DPD and estimating the AoAs of NLoS paths; 2) $\mathbf{CRB}^{\mathrm{J}}_{\mathbf{p}_0\mathbf{p}_0}$ for joint DPD and estimating AoAs and ToAs of all paths. Then we have

$$\mathbf{CRB}^{\mathrm{J}}_{\mathbf{p}_0\mathbf{p}_0} \leq \mathbf{CRB}^{\mathrm{O}}_{\mathbf{p}_0\mathbf{p}_0}. \tag{40}$$

Proof: See Appendix B.

V. Numerical Results

A. Basics and Assumptions

To evaluate the performance of the proposed DIM-DPD, we compare it with ADMM-DPD [42], which is one of the iterative implementations for the sparse DPD problem, and with SM-DPD [39] that is one of the subspace-based methods. The root mean-squared error (RMSE) is adopted as the main performance metric, defined as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N_{Mc}}\sum\nolimits_{n_{Mc}=1}^{N_{Mc}} \left\|\hat{\mathbf{p}}_0^{[n_{Mc}]} - \mathbf{p}_0\right\|^2}, \tag{41}$$

where N_{Mc} is the number of Monte Carlo trials, and $\hat{\mathbf{p}}_0^{[n_{Mc}]}$ is the estimated target position at the n_{Mc}-th trial. All DPD algorithms are also compared with closed-form CRLB, which is derived in Appendix A and reflects the theoretically achievable performance taken as the ultimate benchmark.

As depicted already in Fig. 1, we consider the scenario with $L = 3$ APs, which are located at $\mathbf{u}_1 = [0, 0]$ m, $\mathbf{u}_2 = [15, 0]$ m, and $\mathbf{u}_3 = [0, 15]$ m, respectively. Each AP is equipped with a ULA with $M = 4$ half-wavelength spaced antennas. UE is located at $\mathbf{p}_0 = [6.38, 10.36]$ m and transmits the OFDM signal like a typical Wi-Fi standard IEEE 802.11ac. The spacing and the total number of subcarriers are, respectively, $\Delta f = 312.5$ kHz and $N_0 = 256$. The signal carrier frequency is $f_c = 5.32$ GHz, and the signal propagation speed is $c = 3 \cdot 10^8$ m/s reflecting the speed-of-light. Then the APs obtain CSI through a deconvolution of the known training sequence and pulse shape function [46]. The transmission time of UE to all APs is selected randomly from the interval $[0, 800]$ ns, while the attenuation of the channel of all paths is also randomly generated following the Rayleigh distribution, and the channel phases follow the uniform distribution $U[0, 2\pi]$. Furthermore, the number of Monte Carlo trials $N_{Mc} = 1000$. For each trial, the proposed DIM-DPD has randomly allocated $S = 5$ initialization groups.

B. Results

Since the proposed algorithm is iterative, we first compare the proposed DIM-DPD method with ADMM-DPD and SM-DPD in terms of global localization error, by evaluating the cumulative distribution function (CDF) of $\|\hat{\mathbf{p}}_0^{[n_{Mc}]} - \mathbf{p}_0\|$. The simulation parameters are set as: SNR = 0 dB, $N = N_0/4 = 64$, and observed LoS ToAs of the UE to each AP are 394.32 ns, 655.05 ns, and 538.93 ns, respectively. Assume that each AP also receives signal from a close NLoS path besides the LoS path. The observed ToAs and AoAs of paths in ascending order of the AP indexes are $[407.69\,\mathrm{ns}, 70.29°]$, $[664.77\,\mathrm{ns}, 120.77°]$, and $[548.78\,\mathrm{ns}, -67.20°]$, respectively. Moreover, the third AP receives an extra path with $[544.78\,\mathrm{ns}, -71.70°]$. Note that this is an intractable multipath scenario for the system with the relative small aperture and limited bandwidth. To highlight this in more concrete terms, Fig. 4 shows the space-time spectra of CSIs from all APs based on the traditional 2D FFT. It can be seen that we are not able to distinguish multiple signal paths (LoS and NLoS paths) in the described scenario based on the space-time spectrum which only relies on the angular and delay resolution.

The obtained localization error CDFs are shown in Fig. 5. It can be clearly seen and observed that the position error CDF curve of the proposed DIM-DPD is much steeper than that of the ADMM-DPD and SM-DPD benchmark solutions because of involving ToA in the signal model and parameter estimation. Furthermore, the probability of sub-meter localization error with the proposed DIM-DPD is close to 100%, while only 70% of the trials achieve sub-meter error when using the ADMM-DPD and SM-DPD reference methods. The results in Fig. 5 validate that the proposed iterative algorithm outperforms both the ADMM-DPD and the SM-DPD on the possibility of converging to the global optima.

Next, to demonstrate the convergence speed of the proposed DIM-DPD, Fig. 6 compares the average number of iterations for DIM-DPD and ADMM-DPD against different SNRs while keeping other simulation parameters the same as in Fig. 5. It can be seen that the DIM-DPD requires similar or lower amount of iterations than the ADMM-DPD in almost all SNR regions. Considering further the fact that the iterations of DIM-DPD use the multiple line searches on each AP, which can be operated in parallel, the actual implementation or execution speed of DIM-DPD will be much higher than that of the ADMM-DPD.

Keeping the simulation parameters again the same as those used in Fig. 6, Fig. 7 presents the RMSE of position estimation

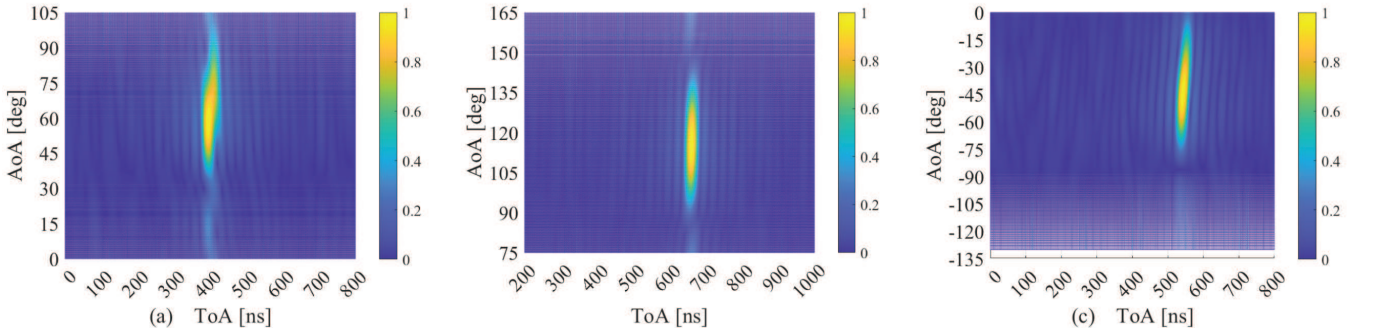


Fig. 4. 2D space-time spectra of CSI extracted from different APs. Case (a) contains LoS path from $[394.32\,\text{ns}, 58.37°]$ and multipath from $[407.69\,\text{ns}, 70.29°]$. Case (b) contains LoS path from $[655.05\,\text{ns}, 110.24°]$ and multipath from $[664.77\,\text{ns}, 120.77°]$. Case (c) contains LoS path from $[538.93\,\text{ns}, -36.03°]$ and multipaths from $[544.78\,\text{ns}, -71.70°]$ and $[548.78\,\text{ns}, -67.20°]$.

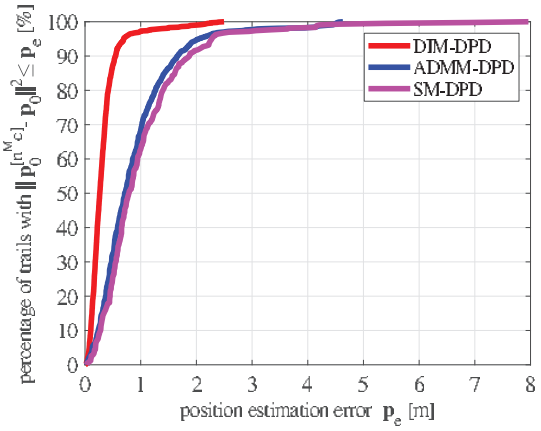


Fig. 5. CDFs of the position estimation errors of the considered DPD algorithms when SNR = 0 dB.

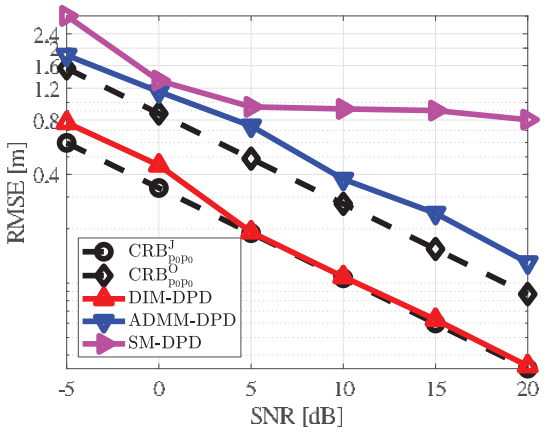


Fig. 7. RMSEs of the DPD algorithms vs. SNR.

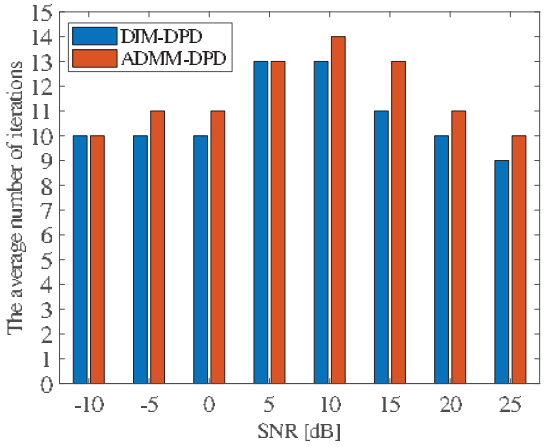


Fig. 6. Comparison of average iteration numbers of iterative ADMM-DPD and the proposed DIM-DPD algorithms vs. SNR.

against SNRs. It can be seen that the proposed DIM-DPD achieves the theoretical performance bound ($\mathbf{CRB}^{\text{J}}_{\mathbf{p}_0\mathbf{p}_0}$) in most SNR regions, while minor deviation is observed only when SNR $\leq$ 0 dB, even if using relatively small aperture and bandwidth limited COTS Wi-Fi devices. The high path resolving capability of the DIM-DPD comes from the fact that it incorporates the observed ToAs and exact model of NLoS paths in the received signal, rather than using the statistical information of multipath as in [38] and [50] or treating the multipath signals as interference or noise [40]. We also observe that the SM-DPD performs poorly because of the lack of the ability to separate the LoS path from the close multipaths. Furthermore, although the ADMM-DPD RMSE shows a consistently decreasing trend similar to $\mathbf{CRB}^{\text{J}}_{\mathbf{p}_0\mathbf{p}_0}$ for increasing SNR, there is a sizable gap to $\mathbf{CRB}^{\text{O}}_{\mathbf{p}_0\mathbf{p}_0}$. The proposed DIM-DPD, in turn, effectively approaches $\mathbf{CRB}^{\text{J}}_{\mathbf{p}_0\mathbf{p}_0}$, particularly when the SNR is greater than 5 dB. Such performance improvement is partially facilitated by the inclusion of the observed ToAs in the iterative parameter estimation procedure. In general, the simulation results match well with the theoretical analysis in Section IV-C, while validating that the proposed DIM-DPD is an accurate implementation of the exact ML estimator at medium and high SNRs.

Besides the SNR, the impact of NLoS paths is also investigated. In Fig. 8, the RMSEs of different algorithms are shown against $|b_{LoS}|/|b_{NLoS}|$ which is the ratio of the relative strengths of LoS and NLoS paths. Setting SNR = 5 dB, Fig. 8 shows that the proposed DIM-DPD always approaches optimal performance $\mathbf{CRB}^{\text{J}}_{\mathbf{p}_0\mathbf{p}_0}$ due to the capability to recognize the LoS path using ToA information. In contrast, both ADMM-DPD and SM-DPD obviously deviate from $\mathbf{CRB}^{\text{O}}_{\mathbf{p}_0\mathbf{p}_0}$. The gap decreases when the power of the LoS path gradually dominates, i.e., for increasing $|b_{LoS}|/|b_{NLoS}|$. Importantly, Fig. 8 indicates that the proposed DIM-DPD algorithm is more robust in scenarios with strong NLoS paths.

TABLE IV
ACCURACY COMPARISON FOR THE PROPOSED DIM-DPD BETWEEN DIFFERENT AP CONFIGURATIONS IN TERMS OF VARYING THE LOCATION OF THE AP3, $\Delta\theta = 10°$

The location of AP3, $\mathbf{u}_3$ (m)	[0,15]	[3,15]	[6,15]	[9,15]	[12,15]	[15,15]
$\mathbf{CRB}^{\mathrm{J}}_{\mathbf{p}_0\mathbf{p}_0}$ **(m)**	0.309	0.25	0.288	0.33	0.357	0.327
RMSE of DIM-DPD (m)	0.315	0.259	0.297	0.336	0.364	0.333

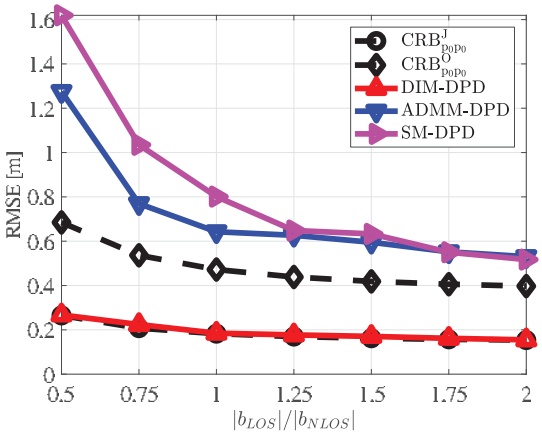


Fig. 8. RMSEs of the compared algorithms vs. $|b_{LoS}|/|b_{NLoS}|$.

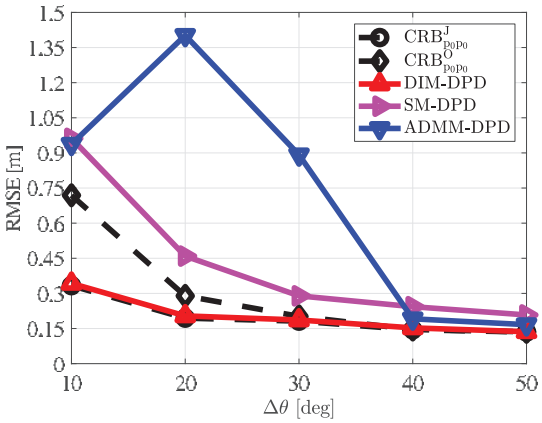


Fig. 9. RMSEs of the compared algorithms vs. AoA separation $\Delta\theta$ between the LoS path and NLoS paths.

Finally, setting SNR = 5 dB and $|b_{LoS}|/|b_{NLoS}| = 1$, the next experiment investigates the impact of the AoA separation $\Delta\theta$ (see Fig. 1) between the LoS and NLoS paths on the DPD estimators. To explicitly show the impact, the number of paths is adjusted to 2 for all APs. Keeping the parameters of all LoS paths unchanged, we present the RMSE for varying $\Delta\theta$ in Fig. 9. It can be seen that DIM-DPD always achieves optimal performance and that its RMSE is highly aligned with $\mathbf{CRB}^{\mathrm{J}}_{\mathbf{p}_0\mathbf{p}_0}$ for all values of $\Delta\theta$. The reason is that the proposed DIM-DPD algorithm implements joint ML estimation for the UE position and all parameters of the NLoS paths. Only when $\Delta\theta \geq 40°$, the RMSE of ADMM-DPD is close to that of DIM-DPD. The deviation of ADMM-DPD even arises at $\Delta\theta = 20°$ due to the neglect of information in the ToAs that potentially helps to separate spatially close NLoS paths from the LoS path. The performance of the sub-optimal SM-DPD gradually approaches $\mathbf{CRB}^{\mathrm{O}}_{\mathbf{p}_0\mathbf{p}_0}$ when $\Delta\theta$ increases; however, there is still a distinct and notable gap. Compared with other algorithms, the results in Fig. 9 indicate that the DIM-DPD has better ability to identify the LoS path, especially when the angle separation between LoS and NLoS is relatively small.

As a concluding study, it is worth to note and highlight that the high-resolution ability of the proposed DIM-DPD method is agnostic to the geometrical configuration of the involved APs. To validate this, at least to an elementary manner, we observe and summarize the RMSEs of the proposed DIM-DPD and the corresponding CRBs for six different AP geometry configurations in Table IV. The AP3 (with location vector $\mathbf{u}_3$) is placed in different locations while keeping $\Delta\theta = 10°$ the same in each configuration. From the results in Table IV, it can be seen that the RMSE of the proposed DIM-DPD stays consistently close to the optimal performance, with only $0.06 \sim 0.09$ m difference from $\mathbf{CRB}^{\mathrm{J}}_{\mathbf{p}_0\mathbf{p}_0}$. This highlights that the proposed DIM-DPD is a generic method, robust to the specific AP deployment and configuration.

VI. CONCLUSION

In this work, we addressed the challenges of accurate device localization with COTS Wi-Fi devices in challenging indoor multipath environments. A decentralized iterative DPD algorithm, called DIM-DPD, was proposed building on the fundamental EM estimation concepts. In each iteration cycle, the estimation of position and nuisance parameters is separated. The first has a closed-form solution, while the second is alternatively estimated. Thus, no multi-dimensional search exists in the proposed algorithm, offering good computational efficiency. Despite inevitable clock biases, the timing components are still introduced into nuisance parameters and are theoretically validated through CRLB, shown to be beneficial for position estimation accuracy. Moreover, only the low-dimensional estimated UE-AP variables are collected from APs to UE, while the communication and concentration of complete CSIs is avoided. Simulation results show the clear superiority of the proposed DIM-DPD over baseline prior-art solutions in terms of position estimation accuracy and communication resource consumption.

APPENDIX A
DERIVATION OF THE CRLB IN CLOSED-FORM

In this appendix, the CRLB is derived for the model (3). First, we define

$$\bar{\mathbf{r}} \triangleq \left[\bar{\mathbf{r}}_1^T, \ldots, \bar{\mathbf{r}}_L^T\right]^T, \quad \mathbf{F} \triangleq \mathrm{blkdiag}\{\mathbf{f}_1, \ldots, \mathbf{f}_L\},$$
$$\bar{\mathbf{F}} \triangleq \mathrm{blkdiag}\{\bar{\mathbf{K}}_1, \ldots, \bar{\mathbf{K}}_L\},$$
$$\bar{\mathbf{K}}_l \triangleq [\mathbf{f}_l, \tilde{\mathbf{F}}_l], \qquad \mathbf{b}_0 \triangleq [b_{10}, \ldots, b_{L0}]^T,$$
$$\tilde{\mathbf{b}} \triangleq [\tilde{\mathbf{b}}_1^T, \ldots, \tilde{\mathbf{b}}_L^T]^T, \quad \bar{\mathbf{w}} \triangleq \left[\bar{\mathbf{w}}_1^T, \ldots, \bar{\mathbf{w}}_L^T\right]^T.$$

Then, the observations from all reception intervals $\bar{\mathbf{r}}$ can be expressed as

$$\bar{\mathbf{r}} = \bar{\mathbf{F}}\bar{\mathbf{b}} + \bar{\mathbf{w}} = \mathbf{F}\mathbf{b}_0 + \mathbf{n} + \bar{\mathbf{w}} \tag{42}$$

where $\mathbf{n} = [(\tilde{\mathbf{F}}_1\tilde{\mathbf{b}}_1)^T, \ldots, (\tilde{\mathbf{F}}_L\tilde{\mathbf{b}}_L)^T]^T$. Respectively concatenating the real and imaginary parts of $\bar{\mathbf{b}}$ is expressed as

$$\bar{\mathbf{b}}_R \triangleq \mathrm{Re}\{\bar{\mathbf{b}}\} \quad \text{and} \quad \bar{\mathbf{b}}_I \triangleq \mathrm{Im}\{\bar{\mathbf{b}}\}. \tag{43}$$

Let $\mathbf{z} = [\mathbf{t}_1^T, \tilde{\boldsymbol{\theta}}_1^T, \ldots, \mathbf{t}_L^T, \tilde{\boldsymbol{\theta}}_L^T]^T$, and the unknown parameter vector $\boldsymbol{\xi}$ can be rewritten as $\boldsymbol{\xi} = [\mathbf{p}_0^T, \mathbf{z}^T, \bar{\mathbf{b}}_R^T, \bar{\mathbf{b}}_I^T]^T$. Then, the CRLB bounds the mean square error of any unbiased estimator of $\boldsymbol{\xi}$, denoted by $\hat{\boldsymbol{\xi}}$, which is commonly expressed as [51]

$$\mathbb{E}\{(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi})(\hat{\boldsymbol{\xi}} - \boldsymbol{\xi})^T\} \geq \mathbf{M}^{-1}(\boldsymbol{\xi}), \tag{44}$$

where $\mathbf{M}(\boldsymbol{\xi})$ is the Fisher information matrix (FIM) of the parameter vector $\boldsymbol{\xi}$. FIM can be partitioned into blocks associated with the different combinations of the components in $\boldsymbol{\xi}$ as

$$\mathbf{M}(\boldsymbol{\xi}) \triangleq \begin{bmatrix} \mathbf{M}_{\mathbf{p}_0\mathbf{p}_0} & \mathbf{M}_{\mathbf{p}_0\mathbf{z}} & \mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}_R} & \mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}_I} \\ \mathbf{M}_{\mathbf{p}_0\mathbf{z}}^T & \mathbf{M}_{\mathbf{z}\mathbf{z}} & \mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}_R} & \mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}_I} \\ \mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}_R}^T & \mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}_R}^T & \mathbf{M}_{\bar{\mathbf{b}}_R\bar{\mathbf{b}}_R} & \mathbf{M}_{\bar{\mathbf{b}}_R\bar{\mathbf{b}}_I} \\ \mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}_I}^T & \mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}_I}^T & \mathbf{M}_{\bar{\mathbf{b}}_R\bar{\mathbf{b}}_I}^T & \mathbf{M}_{\bar{\mathbf{b}}_I\bar{\mathbf{b}}_I} \end{bmatrix}, \tag{45}$$

where $\mathbf{M}_{\mathbf{p}_0\mathbf{p}_0}$ is the FIM associated with the source's position vector $\mathbf{p}_0$.

The CSIs from all APs $\bar{\mathbf{r}}$ is a proper, complex Gaussian, random vector whose mean and covariance are given by

$$\boldsymbol{\chi}^{\mathrm{J}} = \mathbf{F}\mathbf{b}_0 + \mathbf{n} \quad \text{and} \quad \boldsymbol{\sigma}^{\mathrm{J}} = \bar{\sigma}^2\mathbf{I}_{LNM}, \tag{46}$$

where $\bar{\sigma}^2$ is the known noise variance. Then entries of the $\mathbf{M}(\boldsymbol{\xi})$ in this case are given by [51]

$$[\mathbf{M}(\boldsymbol{\xi})]_{g,h} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\left\{\left(\frac{\partial\boldsymbol{\chi}^{\mathrm{J}}(\boldsymbol{\xi})}{\partial[\boldsymbol{\xi}]_g}\right)^H\left(\frac{\partial\boldsymbol{\chi}^{\mathrm{J}}(\boldsymbol{\xi})}{\partial[\boldsymbol{\xi}]_h}\right)\right\}. \tag{47}$$

The derivatives that are necessary for obtaining $\mathbf{M}(\boldsymbol{\xi})$ can be calculated as

$$\frac{\partial\boldsymbol{\chi}^{\mathrm{J}}}{\partial\mathbf{p}_0} = \dot{\mathbf{F}}\bar{\boldsymbol{\alpha}}_{\mathbf{p}_0}, \frac{\partial\boldsymbol{\chi}^{\mathrm{J}}}{\partial\mathbf{z}} = \dot{\bar{\mathbf{F}}}\bar{\boldsymbol{\alpha}}_{\mathbf{z}}, \frac{\partial\boldsymbol{\chi}^{\mathrm{J}}}{\partial\mathbf{b}} = [\bar{\mathbf{F}}, j\bar{\mathbf{F}}], \tag{48}$$

where

$$\dot{\mathbf{F}} \triangleq \left[\frac{\partial\mathbf{F}}{\partial x}, \frac{\partial\mathbf{F}}{\partial y}\right], \bar{\boldsymbol{\alpha}}_{\mathbf{p}_0} \triangleq \mathbf{I}_2 \otimes \mathbf{b}_0, \mathbf{b} \triangleq \left[\bar{\mathbf{b}}_R^T, \bar{\mathbf{b}}_I^T\right]^T$$

$$\dot{\bar{\mathbf{F}}} \triangleq \mathrm{blkdiag}\left\{\left[\dot{\bar{\mathbf{F}}}_1^{\mathrm{t}}, \dot{\bar{\mathbf{F}}}_1^{\theta}\right], \ldots, \left[\dot{\bar{\mathbf{F}}}_L^{\mathrm{t}}, \dot{\bar{\mathbf{F}}}_L^{\theta}\right]\right\}$$

$$\bar{\boldsymbol{\alpha}}_{\mathbf{z}} \triangleq \mathrm{diag}\left\{\left[b_{10}, \tilde{\mathbf{b}}_l^T, \tilde{\mathbf{b}}_l^T, \ldots, b_{L0}, \tilde{\mathbf{b}}_L^T, \tilde{\mathbf{b}}_L^T\right]^T\right\}$$

$$\dot{\bar{\mathbf{F}}}_l^{\mathrm{t}} \triangleq \left[\frac{\partial\mathbf{f}_l}{\partial t_{l0}}, \frac{\mathbf{h}_l(t_{l1})}{\partial t_{l1}} \otimes \mathbf{a}_l\left(\tilde{\theta}_{l1}\right), \ldots, \frac{\mathbf{h}_l(t_{lQ_l})}{\partial t_{lQ_l}} \otimes \mathbf{a}_l\left(\tilde{\theta}_{lQ_l}\right)\right]$$

$$\dot{\bar{\mathbf{F}}}_l^{\theta} \triangleq \left[\mathbf{h}_l(t_{l1}) \otimes \frac{\partial\mathbf{a}_l\left(\tilde{\theta}_{l1}\right)}{\partial\tilde{\theta}_{l1}}, \ldots, \mathbf{h}_l(t_{lQ_l}) \otimes \frac{\partial\mathbf{a}_l\left(\tilde{\theta}_{lQ_l}\right)}{\partial\tilde{\theta}_{lQ_l}}\right].$$

After some manipulations, the submatrices of the FIM (45) can be shown to read as follows

$$\mathbf{M}_{\mathbf{p}_0\mathbf{p}_0} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\bar{\boldsymbol{\alpha}}_{\mathbf{p}_0}^H\left[\dot{\mathbf{F}}^H\dot{\mathbf{F}}\right]\bar{\boldsymbol{\alpha}}_{\mathbf{p}_0}\},$$

$$\mathbf{M}_{\mathbf{z}\mathbf{z}} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\bar{\boldsymbol{\alpha}}_{\mathbf{z}}^H[\dot{\bar{\mathbf{F}}}^H\dot{\bar{\mathbf{F}}}]\bar{\boldsymbol{\alpha}}_{\mathbf{z}}\},$$

$$\mathbf{M}_{\mathbf{p}_0\mathbf{z}} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\bar{\boldsymbol{\alpha}}_{\mathbf{p}_0}^H[\dot{\mathbf{F}}^H\dot{\bar{\mathbf{F}}}]\bar{\boldsymbol{\alpha}}_{\mathbf{z}}\},$$

$$\mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}_R} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\bar{\boldsymbol{\alpha}}_{\mathbf{p}_0}^H[\dot{\mathbf{F}}^H\bar{\mathbf{F}}]\},$$

$$\mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}_I} = -\frac{2}{\bar{\sigma}^2}\mathrm{Im}\{\bar{\boldsymbol{\alpha}}_{\mathbf{p}_0}^H[\dot{\mathbf{F}}^H\bar{\mathbf{F}}]\},$$

$$\mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}_R} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\bar{\boldsymbol{\alpha}}_{\mathbf{z}}^H[\dot{\bar{\mathbf{F}}}^H\bar{\mathbf{F}}]\},$$

$$\mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}_I} = -\frac{2}{\bar{\sigma}^2}\mathrm{Im}\{\bar{\boldsymbol{\alpha}}_{\mathbf{z}}^H[\dot{\bar{\mathbf{F}}}^H\bar{\mathbf{F}}]\},$$

$$\mathbf{M}_{\bar{\mathbf{b}}_R\bar{\mathbf{b}}_R} = \mathbf{M}_{\bar{\mathbf{b}}_I\bar{\mathbf{b}}_I} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\bar{\mathbf{F}}^H\bar{\mathbf{F}}\},$$

$$\mathbf{M}_{\bar{\mathbf{b}}_R\bar{\mathbf{b}}_I} = -\mathbf{M}_{\bar{\mathbf{b}}_I\bar{\mathbf{b}}_R} = -\frac{2}{\bar{\sigma}^2}\mathrm{Im}\{\bar{\mathbf{F}}^H\bar{\mathbf{F}}\}.$$

The position CRLB matrix $\mathbf{CRB}_{\mathbf{p}_0\mathbf{p}_0}^{\mathrm{J}}$ is obtained by the upper left block of the FIM inverse corresponding to $\mathbf{M}_{\mathbf{p}_0\mathbf{p}_0}$. Thus using the partitioned matrix inversion lemma, $\mathbf{CRB}_{\mathbf{p}_0\mathbf{p}_0}^{\mathrm{J}}$ can be given as

$$\begin{aligned}(\mathbf{CRB}_{\mathbf{p}_0\mathbf{p}_0}^{\mathrm{J}})^{-1} &= \mathbf{M}_{\mathbf{p}_0\mathbf{p}_0} - \mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}}\mathbf{M}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}^{-1}\mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}}^T \\ &- (\mathbf{M}_{\mathbf{p}_0\mathbf{z}} - \mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}}\mathbf{M}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}^{-1}\mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}}^T)(\mathbf{M}_{\mathbf{z}\mathbf{z}} - \mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}}\mathbf{M}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}^{-1}\mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}}^T)^{-1} \\ &\cdot (\mathbf{M}_{\mathbf{p}_0\mathbf{z}}^T - \mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}}\mathbf{M}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}^{-1}\mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}}^T),\end{aligned} \tag{49}$$

where

$$\mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}} \triangleq [\mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}_R}, \mathbf{M}_{\mathbf{p}_0\bar{\mathbf{b}}_I}], \mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}} \triangleq [\mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}_R}, \mathbf{M}_{\mathbf{z}\bar{\mathbf{b}}_I}],$$

$$\mathbf{M}_{\bar{\mathbf{b}}\bar{\mathbf{b}}} \triangleq \begin{bmatrix} \mathbf{M}_{\bar{\mathbf{b}}_R\bar{\mathbf{b}}_R} & \mathbf{M}_{\bar{\mathbf{b}}_R\bar{\mathbf{b}}_I} \\ \mathbf{M}_{\bar{\mathbf{b}}_I\bar{\mathbf{b}}_R} & \mathbf{M}_{\bar{\mathbf{b}}_I\bar{\mathbf{b}}_I} \end{bmatrix}.$$

Towards further simplification of (49), we recall the following rules [46]

$$\begin{bmatrix} \mathrm{Re}\{\mathbf{X}\} & -\mathrm{Im}\{\mathbf{X}\} \\ \mathrm{Im}\{\mathbf{X}\} & \mathrm{Re}\{\mathbf{X}\} \end{bmatrix}^{-1} = \begin{bmatrix} \mathrm{Re}\{\mathbf{X}^{-1}\} & -\mathrm{Im}\{\mathbf{X}^{-1}\} \\ \mathrm{Im}\{\mathbf{X}^{-1}\} & \mathrm{Re}\{\mathbf{X}^{-1}\} \end{bmatrix}, \tag{50}$$

and

$$\begin{aligned}\mathrm{Re}\{\mathbf{X}\}\mathrm{Re}\{\mathbf{Y}\} - \mathrm{Im}\{\mathbf{X}\}\mathrm{Im}\{\mathbf{Y}\} &= \mathrm{Re}\{\mathbf{X}\mathbf{Y}\}, \\ \mathrm{Im}\{\mathbf{X}\}\mathrm{Re}\{\mathbf{Y}\} + \mathrm{Re}\{\mathbf{X}\}\mathrm{Im}\{\mathbf{Y}\} &= \mathrm{Im}\{\mathbf{X}\mathbf{Y}\}.\end{aligned} \tag{51}$$

Finally, by using (50) and (51), the expression of $\mathbf{CRB}_{\mathbf{p}_0\mathbf{p}_0}$ can be simplified as

$$\mathbf{CRB}_{\mathbf{p}_0\mathbf{p}_0}^{\mathrm{J}} = \frac{\bar{\sigma}^2}{2}(\boldsymbol{\Gamma}_1 - \boldsymbol{\Gamma}_2\boldsymbol{\Gamma}_3^{-1}\boldsymbol{\Gamma}_2^T)^{-1}, \tag{52}$$

where

$$\begin{aligned}\boldsymbol{\Gamma}_1 &= \mathrm{Re}\{\bar{\boldsymbol{\alpha}}_{\mathbf{p}_0}^H(\dot{\mathbf{F}}^H\mathbf{P}_{\bar{\mathbf{F}}}^{\perp}\dot{\mathbf{F}})\bar{\boldsymbol{\alpha}}_{\mathbf{p}_0}\}, \\ \boldsymbol{\Gamma}_2 &= \mathrm{Re}\{\bar{\boldsymbol{\alpha}}_{\mathbf{p}_0}^H(\dot{\mathbf{F}}^H\mathbf{P}_{\bar{\mathbf{F}}}^{\perp}\dot{\bar{\mathbf{F}}})\bar{\boldsymbol{\alpha}}_{\mathbf{z}}\}, \\ \boldsymbol{\Gamma}_3 &= \mathrm{Re}\{\bar{\boldsymbol{\alpha}}_{\mathbf{z}}^H(\dot{\bar{\mathbf{F}}}^H\mathbf{P}_{\bar{\mathbf{F}}}^{\perp}\dot{\bar{\mathbf{F}}})\bar{\boldsymbol{\alpha}}_{\mathbf{z}}\}, \\ \mathbf{P}_{\bar{\mathbf{F}}}^{\perp} &= \mathbf{I}_{LNM} - \bar{\mathbf{F}}(\bar{\mathbf{F}}^H\bar{\mathbf{F}})^{-1}\bar{\mathbf{F}}^H.\end{aligned} \tag{53}$$

Appendix B
Proof of $\mathbf{CRB}_{\mathbf{p}_0\mathbf{p}_0}^{\mathrm{J}} \leq \mathbf{CRB}_{\mathbf{p}_0\mathbf{p}_0}^{\mathrm{O}}$

To investigate the effect of estimating ToAs on the CRLB of $\hat{\mathbf{p}}_0$, firstly, we rewrite $\boldsymbol{\xi}$ as

$$\boldsymbol{\xi} = [\boldsymbol{\vartheta}^T, \bar{\mathbf{t}}^T, \bar{\mathbf{b}}_R^T, \bar{\mathbf{b}}_I^T]^T. \tag{54}$$

where $\boldsymbol{\vartheta} = [\mathbf{p}_0^T, \bar{\boldsymbol{\theta}}^T]$ and corresponding CRLB is given by

$$\begin{aligned}\left(\mathbf{CRB}^{\mathrm{J}}_{\boldsymbol{\vartheta}\boldsymbol{\vartheta}}\right)^{-1} &= \mathbf{M}_{\boldsymbol{\vartheta}\boldsymbol{\vartheta}} - \mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}}\mathbf{M}^{-1}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}} \\ &\quad - (\mathbf{M}_{\boldsymbol{\vartheta}\mathbf{t}} - \mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}}\mathbf{M}^{-1}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}\mathbf{M}^{T}_{\mathbf{t}\bar{\mathbf{b}}})(\mathbf{M}_{\mathbf{t}\mathbf{t}} - \mathbf{M}_{\mathbf{t}\bar{\mathbf{b}}}\mathbf{M}^{-1}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}\mathbf{M}^{T}_{\mathbf{t}\bar{\mathbf{b}}})^{-1} \\ &\quad \cdot (\mathbf{M}^{T}_{\boldsymbol{\vartheta}\mathbf{t}} - \mathbf{M}_{\mathbf{t}\bar{\mathbf{b}}}\mathbf{M}^{-1}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}}). \end{aligned} \tag{55}$$

On the condition that joint DPD and estimating the AoAs of NLoS paths, the CRLB for estimating $\boldsymbol{\vartheta}$ can be similarly given by

$$\left(\mathbf{CRB}^{\mathrm{O}}_{\boldsymbol{\vartheta}\boldsymbol{\vartheta}}\right)^{-1} = \mathbf{M}_{\boldsymbol{\vartheta}\boldsymbol{\vartheta}} - \mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t}\mathbf{M}^{-1}_{\bar{\mathbf{b}}_t\bar{\mathbf{b}}_t}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t}, \tag{56}$$

where $\bar{\mathbf{b}}_t \triangleq [\mathbf{b}_{t0}^T, \ldots, \mathbf{b}_{tn}^T, \ldots, \mathbf{b}_{t(N-1)}^T]^T$ is the transformed attenuation factor vector while we also further define

$$\tilde{\mathbf{h}}_n \triangleq [\tilde{\mathbf{h}}_{n1}^T, \ldots, \tilde{\mathbf{h}}_{nl}^T, \ldots, \tilde{\mathbf{h}}_{nL}^T]^T, \tag{57}$$

such that

$$\begin{aligned}\tilde{\mathbf{h}}_{nl} &= [e^{-j2\pi n\Delta f t_{l0}}, \ldots, e^{-j2\pi n\Delta f t_{lQ_l}}]^T \\ \mathbf{b}_{tn} &= [\bar{\mathbf{b}}_R^T, \bar{\mathbf{b}}_I^T]^T \odot [\tilde{\mathbf{h}}_n^T, \tilde{\mathbf{h}}_n^T]^T. \end{aligned} \tag{58}$$

Based on (55) and (56), $\mathbf{CRB}^{\mathrm{J}}_{\boldsymbol{\vartheta}\boldsymbol{\vartheta}} \le \mathbf{CRB}^{\mathrm{O}}_{\boldsymbol{\vartheta}\boldsymbol{\vartheta}}$ is sufficient for $\mathbf{CRB}^{\mathrm{J}}_{\mathbf{p}_0\mathbf{p}_0} \le \mathbf{CRB}^{\mathrm{O}}_{\mathbf{p}_0\mathbf{p}_0}$, i.e.,

$$\begin{aligned}&\mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t}\mathbf{M}^{-1}_{\bar{\mathbf{b}}_t\bar{\mathbf{b}}_t}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t} - \mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}}\mathbf{M}^{-1}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}} \\ &\quad - (\mathbf{M}_{\boldsymbol{\vartheta}\mathbf{t}} - \mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}}\mathbf{M}^{-1}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}\mathbf{M}^{T}_{\mathbf{t}\bar{\mathbf{b}}})(\mathbf{M}_{\mathbf{t}\mathbf{t}} - \mathbf{M}_{\mathbf{t}\bar{\mathbf{b}}}\mathbf{M}^{-1}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}\mathbf{M}^{T}_{\mathbf{t}\bar{\mathbf{b}}})^{-1} \\ &\quad \cdot (\mathbf{M}^{T}_{\boldsymbol{\vartheta}\mathbf{t}} - \mathbf{M}_{\mathbf{t}\bar{\mathbf{b}}}\mathbf{M}^{-1}_{\bar{\mathbf{b}}\bar{\mathbf{b}}}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}}) \ge 0. \end{aligned} \tag{59}$$

Using the standard technique of Schur complements, (59) can be transformed as

$$\mathbf{M}_c = \begin{bmatrix} \mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t}\mathbf{M}^{-1}_{\bar{\mathbf{b}}_t\bar{\mathbf{b}}_t}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t} & \mathbf{M}_{\boldsymbol{\vartheta}\mathbf{t}} & \mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}} \\ \mathbf{M}^{T}_{\boldsymbol{\vartheta}\mathbf{t}} & \mathbf{M}_{\mathbf{t}\mathbf{t}} & \mathbf{M}_{\mathbf{t}\bar{\mathbf{b}}} \\ \mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}} & \mathbf{M}^{T}_{\mathbf{t}\bar{\mathbf{b}}} & \mathbf{M}_{\bar{\mathbf{b}}\bar{\mathbf{b}}} \end{bmatrix} \ge 0. \tag{60}$$

Note that the signal model corresponding to $\mathbf{CRB}^{\mathrm{O}}_{\boldsymbol{\vartheta}\boldsymbol{\vartheta}}$ has N snapshots, and the mean and covariance of $(n+1)$-th one are respectively given by

$$\boldsymbol{\chi}^{\mathrm{O}} = \mathbf{A}\mathbf{b}_{tn} \quad \text{and} \quad \boldsymbol{\sigma}^{\mathrm{O}} = \bar{\sigma}^2\mathbf{I}_{LM}, \tag{61}$$

where $\mathbf{A} \triangleq \mathrm{blkdiag}\{\mathbf{A}_1, \ldots, \mathbf{A}_l, \ldots, \mathbf{A}_L\}$ and $\mathbf{A}_l \triangleq [\mathbf{a}_l(\mathbf{p}_0), \mathbf{a}_l(\tilde{\theta}_{l1}), \ldots, \mathbf{a}_l(\tilde{\theta}_{lQ_l})]$. The submatrices in $\mathbf{M}_c$ respectively are

$$\mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t} = [\mathbf{M}_{\boldsymbol{\vartheta}\mathbf{b}_{t0}}, \ldots, \mathbf{M}_{\boldsymbol{\vartheta}\mathbf{b}_{t(N-1)}}],$$

$$\mathbf{M}_{\bar{\mathbf{b}}_t\bar{\mathbf{b}}_t} = \mathrm{blkdiag}\left\{\mathbf{M}_{\mathbf{b}_{t0}\mathbf{b}_{t0}}, \ldots, \mathbf{M}_{\mathbf{b}_{t(N-1)}\mathbf{b}_{t(N-1)}}\right\},$$

$$\mathbf{M}_{\boldsymbol{\vartheta}\mathbf{b}_{tn}} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\mathbf{B}_n^H\dot{\mathbf{A}}^H[\mathbf{A}, j\mathbf{A}]\},$$

$$\mathbf{M}_{\boldsymbol{\vartheta}\mathbf{t}} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\left[\sum\nolimits_{n=0}^{N-1}\mathbf{B}_n^H\dot{\mathbf{A}}^H\mathbf{A}\mathrm{diag}(\tilde{\mathbf{h}}_n')\right]\mathrm{diag}(\bar{\mathbf{b}})\},$$

$$\mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_R} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\sum\nolimits_{n=0}^{N-1}\mathbf{B}_n^H\dot{\mathbf{A}}^H\mathbf{A}\mathrm{diag}(\tilde{\mathbf{h}}_n)\},$$

$$\mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_I} = -\frac{2}{\bar{\sigma}^2}\mathrm{Im}\{\sum\nolimits_{n=0}^{N-1}\mathbf{B}_n^H\dot{\mathbf{A}}^H\mathbf{A}\mathrm{diag}(\tilde{\mathbf{h}}_n)\},$$

$$\mathbf{M}_{\mathbf{t}\mathbf{t}} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\mathrm{diag}(\bar{\mathbf{b}}^c)\big[\sum\nolimits_{n=0}^{N-1}\mathrm{diag}((\tilde{\mathbf{h}}_n')^c)\mathbf{A}^H\cdot \mathbf{A}\mathrm{diag}(\tilde{\mathbf{h}}_n')\big]\mathrm{diag}(\bar{\mathbf{b}})\},$$

$$\mathbf{M}_{\mathbf{t}\bar{\mathbf{b}}_R} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\mathrm{diag}(\bar{\mathbf{b}}^c)\sum\nolimits_{n=0}^{N-1}\mathrm{diag}((\tilde{\mathbf{h}}_n')^c)\mathbf{A}^H\cdot \mathbf{A}\mathrm{diag}(\tilde{\mathbf{h}}_n)\},$$

$$\mathbf{M}_{\mathbf{t}\bar{\mathbf{b}}_I} = -\frac{2}{\bar{\sigma}^2}\mathrm{Im}\{\mathrm{diag}(\bar{\mathbf{b}}^c)\big[\sum\nolimits_{n=0}^{N-1}\mathrm{diag}((\tilde{\mathbf{h}}_n')^c)\mathbf{A}^H\cdot \mathbf{A}\mathrm{diag}(\tilde{\mathbf{h}}_n)\big]\},$$

$$\mathbf{M}_{\bar{\mathbf{b}}_R\bar{\mathbf{b}}_R} = \mathbf{M}_{\bar{\mathbf{b}}_I\bar{\mathbf{b}}_I} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\left[\mathrm{diag}(\tilde{\mathbf{h}}_n^c)\sum\nolimits_{n=0}^{N-1}\mathbf{A}^H\mathbf{A}\mathrm{diag}(\tilde{\mathbf{h}}_n)\right]\},$$

$$\mathbf{M}_{\bar{\mathbf{b}}_R\bar{\mathbf{b}}_I} = \frac{2}{\bar{\sigma}^2}\mathrm{Im}\{\left[\mathrm{diag}(\tilde{\mathbf{h}}_n^c)\sum\nolimits_{n=0}^{N-1}\mathbf{A}^H\mathbf{A}\mathrm{diag}(\tilde{\mathbf{h}}_n)\right]\},$$

$$\mathbf{M}_{\bar{\mathbf{b}}_I\bar{\mathbf{b}}_R} = -\frac{2}{\bar{\sigma}^2}\mathrm{Im}\{\left[\mathrm{diag}(\tilde{\mathbf{h}}_n^c)\sum\nolimits_{n=0}^{N-1}\mathbf{A}^H\mathbf{A}\mathrm{diag}(\tilde{\mathbf{h}}_n)\right]\},$$

where $\dot{\mathbf{A}} = [\dot{\mathbf{A}}_{0x}, \dot{\mathbf{A}}_{0y}, \dot{\mathbf{A}}_{\mathbf{n}}]$, $\mathbf{B}_n = \mathrm{blkdiag}\left\{\mathbf{B}_n^0, \mathbf{B}_n^{\mathbf{n}}\right\}$ and

$$\begin{aligned}\mathbf{M}_{\mathbf{b}_{tn}\mathbf{b}_{tn}} &= \frac{2}{\bar{\sigma}^2}\begin{bmatrix}\mathrm{Re}(\mathbf{A}^H\mathbf{A}) & -\mathrm{Im}(\mathbf{A}^H\mathbf{A}) \\ \mathrm{Im}(\mathbf{A}^H\mathbf{A}) & \mathrm{Re}(\mathbf{A}^H\mathbf{A})\end{bmatrix}, \\ \mathbf{B}_n^0 &= \mathbf{I}_2 \otimes (\mathbf{b}_0 \odot \check{\mathbf{h}}_{n0}), \\ \mathbf{B}_n^{\mathbf{n}} &= \mathrm{diag}\{[(\tilde{\mathbf{b}}_1^T \odot \check{\mathbf{h}}_{n1}^T), \ldots, (\tilde{\mathbf{b}}_L^T \odot \check{\mathbf{h}}_{nL}^T)]^T\}, \\ \dot{\mathbf{A}}_{0x} &= \mathrm{blkdiag}\left\{\frac{\partial\mathbf{a}_1(\mathbf{p}_0)}{\partial x}, \ldots, \frac{\partial\mathbf{a}_L(\mathbf{p}_0)}{\partial x}\right\}, \\ \dot{\mathbf{A}}_{0y} &= \mathrm{blkdiag}\left\{\frac{\partial\mathbf{a}_1(\mathbf{p}_0)}{\partial y}, \ldots, \frac{\partial\mathbf{a}_L(\mathbf{p}_0)}{\partial y}\right\}, \\ \dot{\mathbf{A}}_{\mathbf{n}} &= \mathrm{blkdiag}\{\dot{\mathbf{A}}_{\mathbf{n}1}, \ldots, \dot{\mathbf{A}}_{\mathbf{n}l}, \ldots, \dot{\mathbf{A}}_{\mathbf{n}L}\}, \\ \dot{\mathbf{A}}_{\mathbf{n}l} &\triangleq \left[\frac{\partial\mathbf{a}_l(\tilde{\theta}_{l1})}{\partial\tilde{\theta}_{l1}}, \ldots, \frac{\partial\mathbf{a}_l(\tilde{\theta}_{lQ_l})}{\partial\tilde{\theta}_{lQ_l}}\right]^T, \\ \check{\mathbf{h}}_{n0} &= [e^{-j2\pi n\Delta f t_{10}}, \ldots, e^{-j2\pi n\Delta f t_{L0}}]^T, \\ \check{\mathbf{h}}_{nl} &= [e^{-j2\pi n\Delta f t_{l1}}, \ldots, e^{-j2\pi n\Delta f t_{lQ_l}}]^T, \\ \tilde{\mathbf{h}}_n' &= -j2\pi n\Delta f\tilde{\mathbf{h}}_n. \end{aligned}$$

Recalling rules (50) and (51), $\mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t}\mathbf{M}^{-1}_{\bar{\mathbf{b}}_t\bar{\mathbf{b}}_t}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t}$ has

$$\mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t}\mathbf{M}^{-1}_{\bar{\mathbf{b}}_t\bar{\mathbf{b}}_t}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t} = \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\sum\nolimits_{n=0}^{N-1}\mathbf{B}_n^H\dot{\mathbf{A}}^H\mathbf{P}_{\mathbf{A}}\dot{\mathbf{A}}\mathbf{B}_n\}. \tag{62}$$

where

$$\mathbf{P}_{\mathbf{A}} = \mathbf{A}(\mathbf{A}^H\mathbf{A})^{-1}\mathbf{A}^H = \mathrm{blkdiag}\{\mathbf{P}_{\mathbf{A}_1}, \ldots, \mathbf{P}_{\mathbf{A}_L}\}. \tag{63}$$

Note that the term $\mathbf{B}_n^H\dot{\mathbf{A}}^H$ can also be treated as the summation of L terms corresponding to different APs. Then we can rewrite (62) as

$$\begin{aligned}\mathbf{M}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t}\mathbf{M}^{-1}_{\bar{\mathbf{b}}_t\bar{\mathbf{b}}_t}\mathbf{M}^{T}_{\boldsymbol{\vartheta}\bar{\mathbf{b}}_t} &= \sum_{l=1}^{L}\mathbf{M}_{\boldsymbol{\vartheta}_l\bar{\mathbf{b}}_t}\mathbf{M}^{-1}_{\bar{\mathbf{b}}_t\bar{\mathbf{b}}_t}\mathbf{M}^{T}_{\boldsymbol{\vartheta}_l\bar{\mathbf{b}}_t}, \\ &= \frac{2}{\bar{\sigma}^2}\mathrm{Re}\{\sum_{n=0}^{N-1}\sum_{l=1}^{L}\mathbf{B}_{nl}^H\dot{\mathbf{A}}_l^H\mathbf{P}_{\mathbf{A}}\dot{\mathbf{A}}_l\mathbf{B}_{nl}\}. \end{aligned} \tag{64}$$

where $\dot{\mathbf{A}}_l$ is constructed by only preserving elements related to l-th AP in $\dot{\mathbf{A}}$ and all other elements are zero. With the same operation, $\bar{\mathbf{A}}_l$, $\widehat{\mathbf{b}_l}$ and $\widehat{\mathbf{h}_{nl}}$ are generated from $\mathbf{A}$, $\tilde{\mathbf{b}}$ and $[\check{\mathbf{h}}_{n1}^T, \ldots, \check{\mathbf{h}}_{nl}^T, \ldots, \check{\mathbf{h}}_{nL}^T]^T$, respectively. Let $\check{\mathbf{b}}_l = [b_{l0}, b_{l0}, \widehat{\mathbf{b}_l}^T]^T$ and $\bar{\mathbf{h}}_{nl} = [e^{-j2\pi n\Delta f t_{l0}}, e^{-j2\pi n\Delta f t_{l0}}, \widehat{\mathbf{h}_{nl}}]$, then we have $\mathbf{B}_{nl} = \mathrm{diag}\{\check{\mathbf{b}}_l \odot \bar{\mathbf{h}}_{nl}\}$ and $\mathbf{P}_{\bar{\mathbf{A}}_l} = \bar{\mathbf{A}}_l(\bar{\mathbf{A}}_l^H\bar{\mathbf{A}}_l)^{-1}\bar{\mathbf{A}}_l^H$.

The same transformation as (64) can also be applied to other blocks in $\mathbf{M}_c$. Thus, $\mathbf{M}_c$ can be divided as $\mathbf{M}_c = \sum_{l=1}^{L} \mathbf{M}_{cl}$ and $\mathbf{M}_{cl}$ is given by

$$\mathbf{M}_{cl} = \frac{2}{\bar{\sigma}^2} \mathrm{Re}\left\{ \mathbf{M}_{l1} \odot \mathbf{M}_{l2} \odot \mathbf{M}_{l3} \right\}, \tag{65}$$

where

$$\mathbf{M}_{l1} = \begin{bmatrix} \dot{\mathbf{A}}_l^H \mathbf{P}_{\bar{\mathbf{A}}_l} \dot{\mathbf{A}}_l & -j\dot{\mathbf{A}}_l^H \bar{\mathbf{A}}_l & \dot{\mathbf{A}}_l^H \bar{\mathbf{A}}_l & j\dot{\mathbf{A}}_l^H \bar{\mathbf{A}}_l \\ j\bar{\mathbf{A}}_l^H \dot{\mathbf{A}}_l & \bar{\mathbf{A}}_l^H \bar{\mathbf{A}}_l & j\bar{\mathbf{A}}_l^H \bar{\mathbf{A}}_l & -\mathbf{A}_l^H \bar{\mathbf{A}}_l \\ \bar{\mathbf{A}}_l^H \dot{\mathbf{A}}_l & -j\bar{\mathbf{A}}_l^H \bar{\mathbf{A}}_l & \bar{\mathbf{A}}_l^H \bar{\mathbf{A}}_l & j\bar{\mathbf{A}}_l^H \bar{\mathbf{A}}_l \\ -j\bar{\mathbf{A}}_l^H \dot{\mathbf{A}}_l & -\bar{\mathbf{A}}_l^H \bar{\mathbf{A}}_l & -j\bar{\mathbf{A}}_l^H \bar{\mathbf{A}}_l & \bar{\mathbf{A}}_l^H \bar{\mathbf{A}}_l \end{bmatrix},$$

$$\mathbf{M}_{l2} = \sum\nolimits_{n=0}^{N-1} \begin{bmatrix} \mathbf{G}_{l1} & \mathbf{G}_{l2} & \mathbf{G}_{l3} & \mathbf{G}_{l3} \\ \mathbf{G}_{l2}^H & \mathbf{G}_4 & \mathbf{G}_5 & \mathbf{G}_5 \\ \mathbf{G}_{l3}^H & \mathbf{G}_5^H & \mathbf{G}_6 & \mathbf{G}_6 \\ \mathbf{G}_{l3}^H & \mathbf{G}_5^H & \mathbf{G}_6^H & \mathbf{G}_6 \end{bmatrix},$$

$$\mathbf{G}_{l1} = \bar{\mathbf{h}}_{nl}^c \bar{\mathbf{h}}_{nl}^T, \mathbf{G}_{l2} = 2\pi n \Delta f \bar{\mathbf{h}}_{nl}^c \tilde{\mathbf{h}}_n^T, \mathbf{G}_{l3} = \bar{\mathbf{h}}_{nl}^c \tilde{\mathbf{h}}_n^T,$$

$$\mathbf{G}_4 = (2\pi n \Delta f)^2 \tilde{\mathbf{h}}_n^c \tilde{\mathbf{h}}_n^T, \mathbf{G}_5 = 2\pi n \Delta f \tilde{\mathbf{h}}_n^c \tilde{\mathbf{h}}_n^T, \mathbf{G}_6 = \tilde{\mathbf{h}}_n^c \tilde{\mathbf{h}}_n^T,$$

$$\mathbf{M}_{l3} = \begin{bmatrix} \check{\mathbf{b}}_l^c \check{\mathbf{b}}_l^T & \check{\mathbf{b}}_l^c \bar{\mathbf{b}}^T & \check{\mathbf{b}}_l^c \mathbf{1}^T & \check{\mathbf{b}}_l^c \mathbf{1}^T \\ \bar{\mathbf{b}}^c \check{\mathbf{b}}_l^T & \bar{\mathbf{b}}^c \bar{\mathbf{b}}^T & \bar{\mathbf{b}}^c \mathbf{1}^T & \bar{\mathbf{b}}^c \mathbf{1}^T \\ \mathbf{1}\check{\mathbf{b}}_l^T & \mathbf{1}\bar{\mathbf{b}}^T & \mathbf{1}\mathbf{1}^T & \mathbf{1}\mathbf{1}^T \\ \mathbf{1}\check{\mathbf{b}}_l^T & \mathbf{1}\bar{\mathbf{b}}^T & \mathbf{1}\mathbf{1}^T & \mathbf{1}\mathbf{1}^T \end{bmatrix}.$$

Note that $\mathbf{1}$ is a vector having the same dimension as $\bar{\mathbf{b}}$ with all of its elements being equal to one. Now proving $\mathbf{M}_{cl} \geq 0$ is sufficient for the proof of $\mathbf{CRB}^{\mathrm{J}}_{\mathbf{p}_0\mathbf{p}_0} \leq \mathbf{CRB}^{\mathrm{o}}_{\mathbf{p}_0\mathbf{p}_0}$.

$\mathbf{M}_{l1}$, $\mathbf{M}_{l2}$ and $\mathbf{M}_{l3}$ can be respectively rewritten as

$$\mathbf{M}_{l1} = \begin{bmatrix} \dot{\mathbf{A}}_l^H \\ j\bar{\mathbf{A}}_l^H \\ \bar{\mathbf{A}}_l^H \\ -j\bar{\mathbf{A}}_l^H \end{bmatrix} \mathbf{P}_{\bar{\mathbf{A}}_l} [\dot{\mathbf{A}}_l, -j\bar{\mathbf{A}}_l, \bar{\mathbf{A}}_l, j\bar{\mathbf{A}}_l],$$

$$\mathbf{M}_{l2} = \sum_{n=0}^{N-1} \begin{bmatrix} \bar{\mathbf{h}}_{nl}^c \\ 2\pi n \Delta f \tilde{\mathbf{h}}_n^c \\ \tilde{\mathbf{h}}_n^c \\ \tilde{\mathbf{h}}_n^c \end{bmatrix} [\bar{\mathbf{h}}_{nl}^T, 2\pi n \Delta f \tilde{\mathbf{h}}_n^T, \tilde{\mathbf{h}}_n^T, \tilde{\mathbf{h}}_n^T],$$

$$\mathbf{M}_{l3} = \begin{bmatrix} \check{\mathbf{b}}_l^c \\ \bar{\mathbf{b}}^c \\ \mathbf{1} \\ \mathbf{1} \end{bmatrix} [\check{\mathbf{b}}_l^T, \bar{\mathbf{b}}^T, \mathbf{1}^T, \mathbf{1}^T]. \tag{66}$$

It can be easily observed from (66) that $\mathbf{M}_{l1}$, $\mathbf{M}_{l2}$, $\mathbf{M}_{l3}$ are all positive semidefinite, i.e.,

$$\mathbf{M}_{l1} \odot \mathbf{M}_{l2} \odot \mathbf{M}_{l3} \geq 0. \tag{67}$$

Equation (67) validates $\mathbf{M}_{cl} \geq 0$. Then, conditions $\mathbf{M}_c \geq 0$ and (59) are further met, thus the proof of $\mathbf{CRB}^{\mathrm{J}}_{\mathbf{p}_0\mathbf{p}_0} \leq \mathbf{CRB}^{\mathrm{o}}_{\mathbf{p}_0\mathbf{p}_0}$ is completed.

REFERENCES

[1] M. Kotaru, P. Zhang, and S. Katti, "Localizing low-power backscatter tags using commodity WiFi," in *Proc. 13th Int. Conf. Emerg. Netw. EXperiments Technol.*, Nov. 2017, pp. 251–262.

[2] M. Kotaru and S. Katti, "Position tracking for virtual reality using commodity WiFi," in *Proc. IEEE CVPR*, Jun. 2017, pp. 68–78.

[3] Y. Gu, X. Zhang, Z. Liu, and F. Ren, "BeSense: Leveraging WiFi channel data and computational intelligence for behavior analysis," *IEEE Comput. Intell. Mag.*, vol. 14, no. 4, pp. 31–41, Nov. 2019.

[4] L. Xie, S. Song, Y. C. Eldar, and K. B. Letaief, "Collaborative sensing in perceptive mobile networks: Opportunities and challenges," *IEEE Wireless Commun.*, vol. 30, no. 1, pp. 16–23, Feb. 2023.

[5] M. Xiao et al., "Millimeter wave communications for future mobile networks," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 9, pp. 1909–1935, Sep. 2017.

[6] J. Palacios, P. Casari, H. Assasa, and J. Widmer, "LEAP: Location estimation and predictive handover with consumer-grade mmWave devices," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, Apr. 2019, pp. 2377–2385.

[7] R. Wang, Z. Xing, E. Liu, and J. Wu, "Joint localization and communication study for intelligent reflecting surface aided wireless communication system," *IEEE Trans. Commun.*, vol. 71, no. 5, pp. 3024–3042, May 2023.

[8] A. Blanco, P. J. Mateo, F. Gringoli, and J. Widmer, "Augmenting mmWave localization accuracy through sub-6 GHz on off-the-shelf devices," in *Proc. 20th Annu. Int. Conf. Mobile Syst., Appl. Services*, Jun. 2022, pp. 477–490.

[9] Y. Zou, W. Liu, K. Wu, and L. M. Ni, "Wi-Fi radar: Recognizing human behavior with commodity Wi-Fi," *IEEE Commun. Mag.*, vol. 55, no. 10, pp. 105–111, Oct. 2017.

[10] Y. He, Y. Chen, Y. Hu, and B. Zeng, "WiFi vision: Sensing, recognition, and detection with commodity MIMO-OFDM WiFi," *IEEE Internet Things J.*, vol. 7, no. 9, pp. 8296–8317, Sep. 2020.

[11] M. A. Youssef, A. Agrawala, and A. U. Shankar, "WLAN location determination via clustering and probability distributions," in *Proc. 1st IEEE Int. Conf. Pervasive Comput. Commun., (PerCom)*, Mar. 2003, pp. 143–150.

[12] F. Adib, Z. Kabelac, and D. Katabi, "Multi-person localization via RF body reflections," in *Proc. 12th USENIX Conf. Netw. Syst. Des. Implement.*, 2015, pp. 279–292.

[13] Z. Wang, L. Xie, and Q. Wan, "Beamspace joint azimuth, elevation, and delay estimation for large-scale MIMO-OFDM system," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.

[14] A. Bose and C. Heng Foh, "A practical path loss model for indoor WiFi positioning enhancement," in *Proc. 6th Int. Conf. Inf., Commun. Signal Process.*, Dec. 2007, pp. 1–5.

[15] M. Youssef and A. Agrawala, "The Horus WLAN location determination system," in *Proc. ACM MobiSys*, 2005, pp. 205–218.

[16] A. Gaber and A. Omar, "A study of wireless indoor positioning based on joint TDOA and DOA estimation using 2-D matrix pencil algorithms and IEEE 802.11ac," *IEEE Trans. Wireless Commun.*, vol. 14, no. 5, pp. 2440–2454, May 2015.

[17] Y. Zeng, D. Wu, J. Xiong, E. Yi, R. Gao, and D. Zhang, "FarSense: Pushing the range limit of WiFi-based respiration sensing with CSI ratio of two antennas," *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 3, no. 3, pp. 1–26, Sep. 2019.

[18] Y. Pan, S. De Bast, and S. Pollin, "Indoor direct positioning with imperfect massive MIMO array using measured near-field channels," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021.

[19] M. Kotaru, K. Joshi, D. Bharadia, and S. Katti, "Spotfi: Decimeter level localization using wifi," in *Proc. ACM SIGCOMM*, 2015, pp. 269–282.

[20] D. Vasisht, S. Kumar, and D. Katabi, "Decimeter-level localization with a single WiFi access point," in *Proc. 13th USENIX Symp. Netw. Syst. Design Implement. (NSDI)*, 2016, pp. 165–178.

[21] K. Jiokeng, G. Jakllari, A. Tchana, and A.-L. Beylot, "When FTM discovered MUSIC: Accurate WiFi-based ranging in the presence of multipath," in *Proc. IEEE INFOCOM - IEEE Conf. Comput. Commun.*, Jul. 2020, pp. 1857–1866.

[22] J. Xiong and K. Jamieson, "Arraytrack: A fine-grained indoor location system," in *Proc. ACM NSDI*, 2013, pp. 71–84.

[23] X. Li, S. Li, D. Zhang, J. Xiong, Y. Wang, and H. Mei, "Dynamic-MUSIC: Accurate device-free indoor localization," in *Proc. ACM Int. Joint Conf. Pervasive Ubiquitous Comput.*, Sep. 2016, pp. 196–207.

[24] E. Soltanaghaei, A. Kalyanaraman, and K. Whitehouse, "Multipath triangulation: Decimeter-level WiFi localization and orientation with a single unaided receiver," in *Proc. 16th Annu. Int. Conf. Mobile Syst., Appl., Services*, Jun. 2018, pp. 376–388.

[25] A. B. Pizarro, J. P. Beltrán, M. Cominelli, F. Gringoli, and J. Widmer, "Accurate ubiquitous localization with off-the-shelf IEEE 802.11ac devices," in *Proc. 19th Annu. Int. Conf. Mobile Syst., Appl., Services*, Jun. 2021, pp. 241–254.

[26] H. Zhao, N. Zhang, and Y. Shen, "Beamspace direct localization for large-scale antenna array systems," *IEEE Trans. Signal Process.*, vol. 68, pp. 3529–3544, 2020.

[27] F. Ma, Z.-M. Liu, and F. Guo, "Direct position determination in asynchronous sensor networks," *IEEE Trans. Veh. Technol.*, vol. 68, no. 9, pp. 8790–8803, Sep. 2019.

[28] A. Amar and A. J. Weiss, "Localization of narrowband radio emitters based on Doppler frequency shifts," *IEEE Trans. Signal Process.*, vol. 56, no. 11, pp. 5500–5508, Nov. 2008.
[29] F. Ma, Z.-M. Liu, and F. Guo, "Distributed direct position determination," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 14007–14012, Nov. 2020.
[30] Z. Wang, B. Tan, E. S. Lohan, L. Ni, M. Valkama, and Q. Wan, "Majorization-minimization-based direct localization using one-bit channel measurements," *IEEE Wireless Commun. Lett.*, vol. 13, no. 5, pp. 1389–1393, May 2024.
[31] A. J. Weiss, "Direct position determination of narrowband radio transmitters," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, vol. 2, 2004, pp. 1–4.
[32] Z. Wang, K. Hao, Y. Sun, L. Xie, and Q. Wan, "A computationally efficient direct position determination algorithm based on OFDM system," *IEEE Commun. Lett.*, vol. 27, no. 3, pp. 841–845, Mar. 2023.
[33] A. J. Weiss and A. Amar, "Direct position determination of multiple radio signals," *EURASIP J. Adv. Signal Process.*, vol. 2005, no. 1, pp. 1–13, Dec. 2005.
[34] T. Tirer and A. J. Weiss, "High resolution direct position determination of radio frequency sources," *IEEE Signal Process. Lett.*, vol. 23, no. 2, pp. 192–196, Feb. 2016.
[35] L. Tzafri and A. J. Weiss, "High-resolution direct position determination using MVDR," *IEEE Trans. Wireless Commun.*, vol. 15, no. 9, pp. 6449–6461, Sep. 2016.
[36] K. Hao and Q. Wan, "Importance sampling based direct maximum likelihood position determination of multiple emitters using finite measurements," *Signal Process.*, vol. 186, Sep. 2021, Art. no. 108111.
[37] Z. Wang, Y. Sun, Q. Wan, L. Xie, and N. Liu, "A modest power consumption maximum likelihood direct position determination approach for multiple targets with moving sensor arrays," *IEEE Sensors J.*, vol. 22, no. 22, pp. 21885–21898, Nov. 2022.
[38] O. Bialer, D. Raphaeli, and A. J. Weiss, "Maximum-likelihood direct position estimation in dense multipath," *IEEE Trans. Veh. Technol.*, vol. 62, no. 5, pp. 2069–2079, Jun. 2013.
[39] J. Bai, G. Wei, S. Wang, X. Wang, and Z. Fei, "Efficient direct localization of OFDM emitters in multipath environment with mobile receivers," *IEEE Trans. Veh. Technol.*, vol. 71, no. 1, pp. 545–556, Jan. 2022.
[40] N. Garcia, A. M. Haimovich, J. A. Dabin, M. Coulon, and M. Lops, "Direct localization of emitters using widely spaced sensors in multipath environments," in *Proc. 48th Asilomar Conf. Signals, Syst. Comput.*, Nov. 2014, pp. 695–700.
[41] N. Garcia, H. Wymeersch, E. G. Larsson, A. M. Haimovich, and M. Coulon, "Direct localization for massive MIMO," *IEEE Trans. Signal Process.*, vol. 65, no. 10, pp. 2475–2487, May 2017.
[42] W. Fan, X. Li, G. Yang, C. Li, and Y. Huang, "Low-complexity mobile user tracking in quantized mmWave MIMO systems," *IEEE Trans. Mobile Comput.*, vol. 23, no. 9, pp. 8569–8581, Sep. 2024.
[43] K. Hao and Q. Wan, "Sparse Bayesian inference-based direct off-grid position determination in multipath environments," *IEEE Wireless Commun. Lett.*, vol. 10, no. 6, pp. 1148–1152, Jun. 2021.
[44] W. Fan, S. Liu, C. Li, and Y. Huang, "Fast direct localization for millimeter wave MIMO systems via deep ADMM unfolding," *IEEE Wireless Commun. Lett.*, vol. 12, no. 4, pp. 748–752, Apr. 2023.
[45] Z. Wang, Y. Sun, L. Xie, N. Liu, and Q. Wan, "An iterative direct position determination approach based on Doppler frequency shifts," *IEEE Trans. Veh. Technol.*, vol. 73, no. 2, pp. 2431–2443, Feb. 2024.
[46] F. Wen, P. Liu, H. Wei, Y. Zhang, and R. C. Qiu, "Joint azimuth, elevation, and delay estimation for 3-D indoor localization," *IEEE Trans. Veh. Technol.*, vol. 67, no. 5, pp. 4248–4261, May 2018.
[47] E. Tzoreff and A. J. Weiss, "Expectation-maximization algorithm for direct position determination," *Signal Process.*, vol. 133, pp. 32–39, Apr. 2017.
[48] M. Oispuu and U. Nickel, "Direct detection and position determination of multiple sources with intermittent emission," *Signal Process.*, vol. 90, no. 12, pp. 3056–3064, Dec. 2010.
[49] A. Azevedo-Filho and R. D. Shachter, "Laplace's method approximations for probabilistic inference in belief networks with continuous variables," in *Proc. 10th Conf. Uncertainty Artif. Intell.*, 1994, pp. 28–36.
[50] O. Bialer, D. Raphaeli, and A. J. Weiss, "Efficient time of arrival estimation algorithm achieving maximum likelihood performance in dense multipath," *IEEE Trans. Signal Process.*, vol. 60, no. 3, pp. 1241–1252, Mar. 2012.
[51] H. L. Van Trees, *Detection, Estimation, and Modulation Theory, Part I: Detection, Estimation, and Linear Modulation Theory*. Hoboken, NJ, USA: Wiley, 2004.



Ziqiang Wang (Member, IEEE) received the B.Sc. and Ph.D. degrees from the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu, in 2016 and 2024, respectively. From October 2022 to May 2024, he was a Visiting Researcher with Tampere University, Finland. His current research interests include wireless localization, integrated sensing and communication (ISAC), and array signal processing. He is a reviewer of several conferences and journals.



Bo Tan (Member, IEEE) is an Associate Professor of communications engineering and signal processing with the Department of Electronic and Electrical Engineering, University College London (UCL). Before joining UCL, he was an Assistant Professor and then an Associate Professor with Tampere University, Finland, where he holds a Docent. His research focuses on radio signal processing and applications in wireless communications and sensing. He also works on the wireless networks and machine learning. Prior to faculty, he conducted multiple post-doctoral research fellow positions in radar and applications in healthcare. He is the Vice Chair of the IEEE Finland AP/ED/MTT Chapter, an Associate Editor of IEEE WIRELESS COMMUNICATIONS LETTERS, and a reviewer of multiple IEEE/ACM/IET journals.



Mikko Valkama (Fellow, IEEE) received the M.Sc. (Tech.) and D.Sc. (Tech.) degrees (Hons.) from Tampere University of Technology, Finland, in 2000 and 2001, respectively. In 2003, he was with the Communications Systems and Signal Processing Institute, SDSU, San Diego, CA, USA, as a Visiting Research Fellow. Currently, he is a Full Professor and the Head of the Unit of Electrical Engineering, Tampere University, Finland. His general research interests include radio communications, radio localization, and radio-based sensing, with particular emphasis on 5G and 6G mobile radio networks.



Lei Xie (Member, IEEE) received the B.E. and Ph.D. degrees from the University of Electronic Science and Technology of China, Chengdu, China, in 2016 and 2021, respectively. He was a Post-Doctoral Fellow with the Department of Electronic and Computer Engineering, The Hong Kong University of Science and Technology, Hong Kong. He is currently an Associate Professor with the School of Cyber Science and Engineering, Southeast University, Nanjing, China. His research interests include integrated sensing and communications, perceptive mobile networks, statistical signal processing, optimization theory, and artificial intelligence. He served as the Tutorial Program Co-Chair for 2025 IEEE 102nd Vehicular Technology Conference Workshop.



Qun Wan (Senior Member, IEEE) received the B.Sc. degree in electronic engineering from Nanjing University, Nanjing, China, in 1993, and the M.Sc. and Ph.D. degrees in electronic engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1996 and 2001, respectively. From 2001 to 2003, he held a post-doctoral position with the Department of Electronic Engineering, Tsinghua University, Beijing, China. Since 2004, he has been a Professor with the Department of Electronic Engineering, UESTC. His research interests include direction finding, radio localization, and radar signal processing.